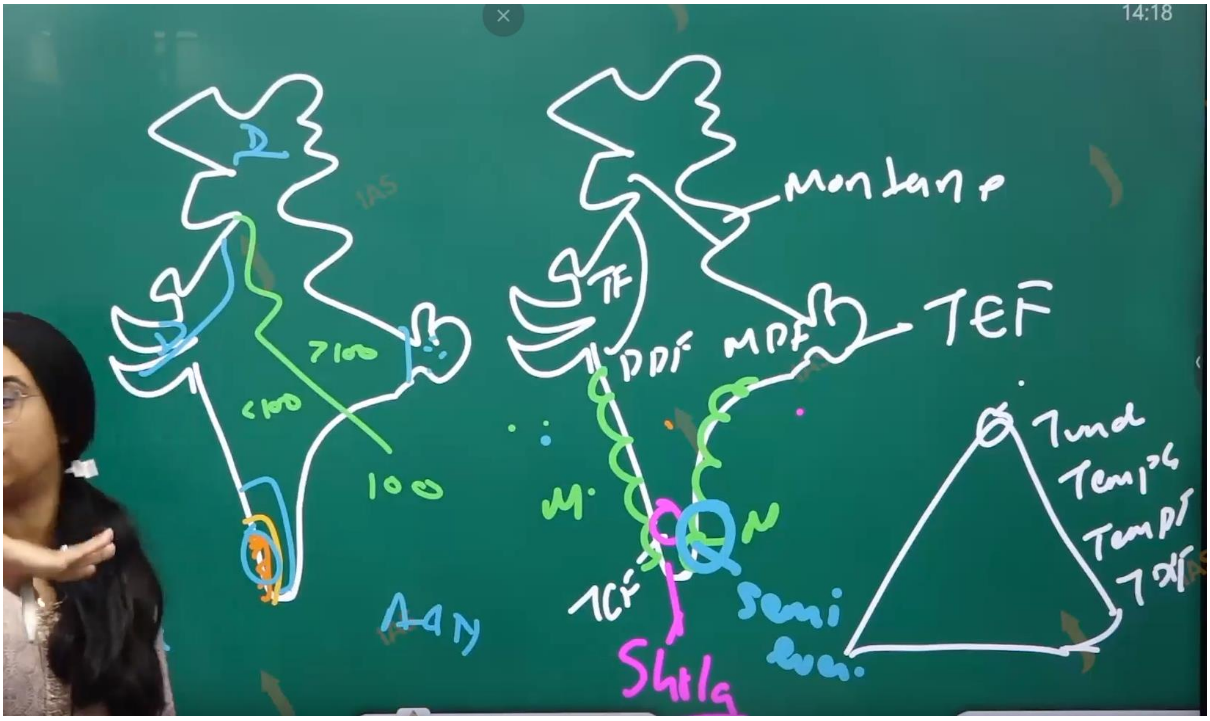
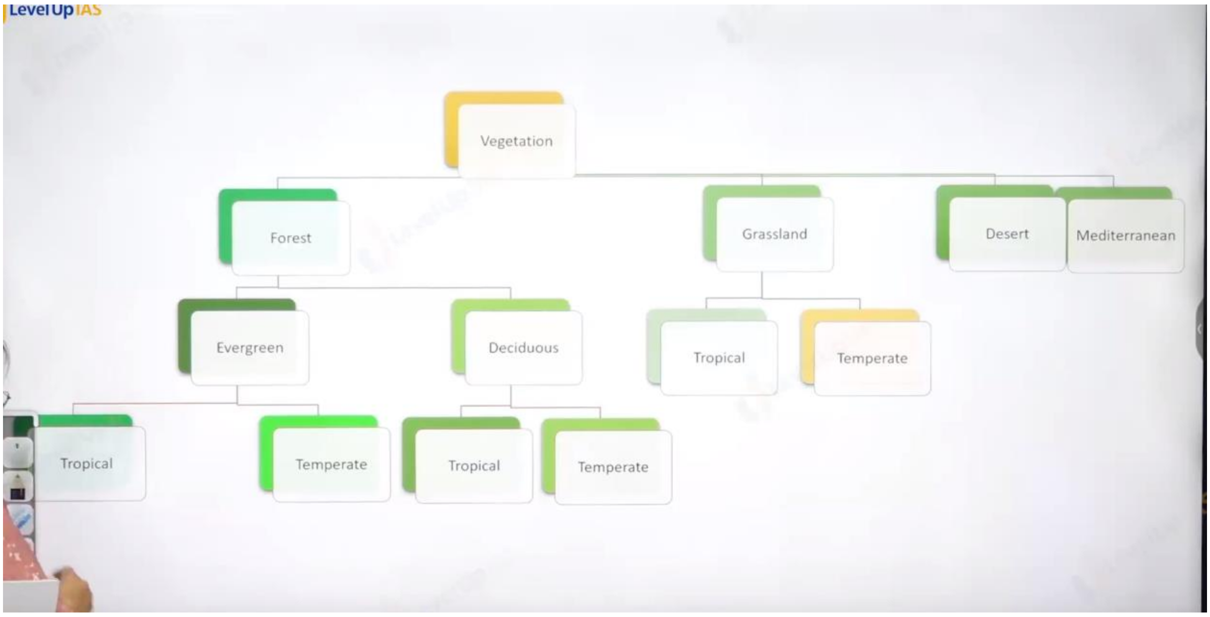
(34) Vegetation

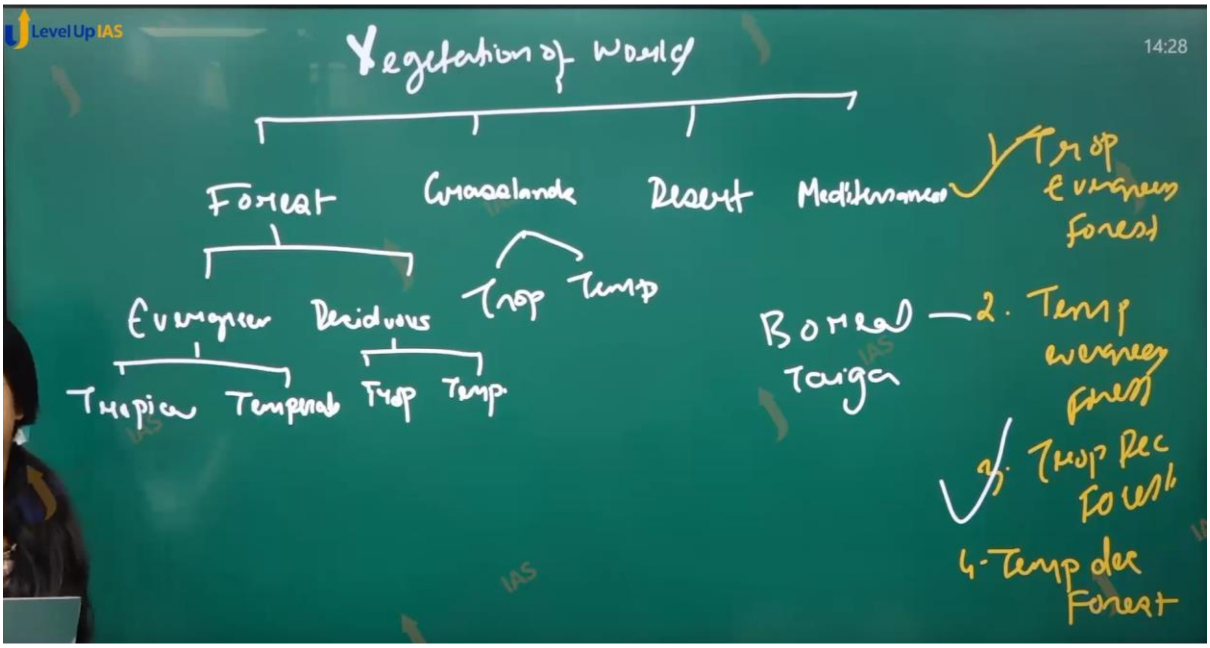



Topic –
Vegetation of
India and Soil
of India
Dimple Nankani

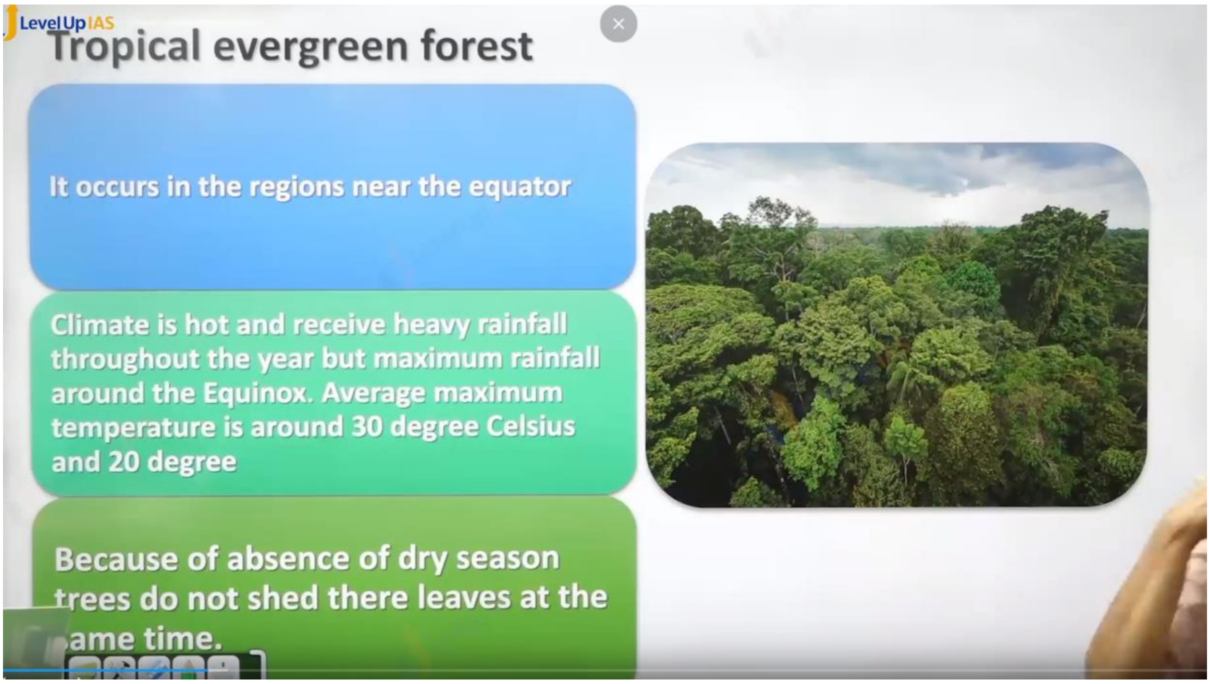

Vegetation
Forest
Grassland
Desert
Mediterranean
Evergreen
Deciduous
Tropical
Temperate
cal
Temperate
Tropical
Temperate


LevelUp IAS
Vegetation
Forest
Grassland
Desert
Mediterranean
Evergreen
Deciduous
Tropical
Temperate
Tropical
Temperate
Tropical
Temperate


evergreen
Tropical
Moscow
Turkestan
Gobi desert
Mohave desert
Sahara desert
Iranian
Arabian desert
Thar desert
Kuala Lumpur
Borneo
Nigeria
Congo
Bogota
Amazon
TDF
Atacama desert
Namib desert
Kalahari desert
Cape town
Australian desert
Perth
Bunbury
Melbourne
Patagonian desert


Level Up IAS
14:28
Vegetation of world
Forest
Grasslands
Desert
Mediterranean
Evergreen
Deciduous
Trop
Temp
Tropical
Temperate
Trop
Temp.
Trop evergreen forest
Boreal
Taiga
2. Temp evergreen forest
3. Trop dec forest
4. Temp dec forest


Level Up IAS
Types of Vegetations
Tropical evergreen forest
Temperate evergreen forest
Tropical deciduous forest
Temperate deciduous forest
Temperate grassland
Tropical grassland
Mediterranean vegetation
Desert vegetation
Conifer forest



Tropical evergreen forest:

---

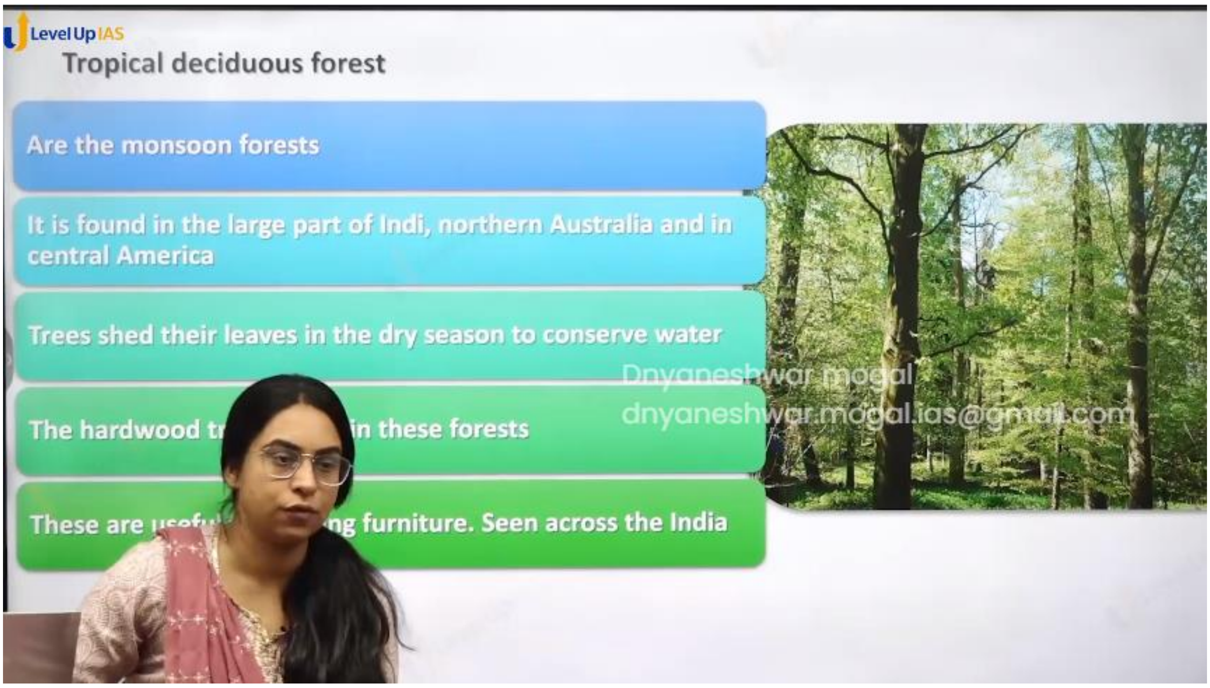
1. It occurs that the region near the equator climate is hot and therefore it receives heavy rainfall throughout the year. But maximum rainfall occurred around the equinox.
2. There is absence of dray season therefore trees do not shade their leaves at same time in the season.
3. The Season of place can be described as each day the same more or less Morning is clear and bright and by afternoon sun climbs up in the sky heating the region forming dark cloud. Leading to rainfall and lightning.
4. It seen in the India Andaman and Nicobar Islands Lakshadweep and southwest side of western ghat, and northeast regions.
5. The soil of tropical evergreen forest is not fertile. As the nutrients are leached down by the rainwater which are then acquired by the plants
6. And soil does not have the major nutrients.
7. This tropical ever green forest once cut cannot be received back.

8. Ex. cutting in Southeast Asia for palm cultivation, cutting in amazon in Brazil for coffee plantations, cutting in Congo forest, for cocoa plantation.
9. The names of species are rosewood, ebony, mahogany. The forest has mix of tree species which not easy to cut them. Some of trees are quobreak tree.

Temperate evergreen forest:

---

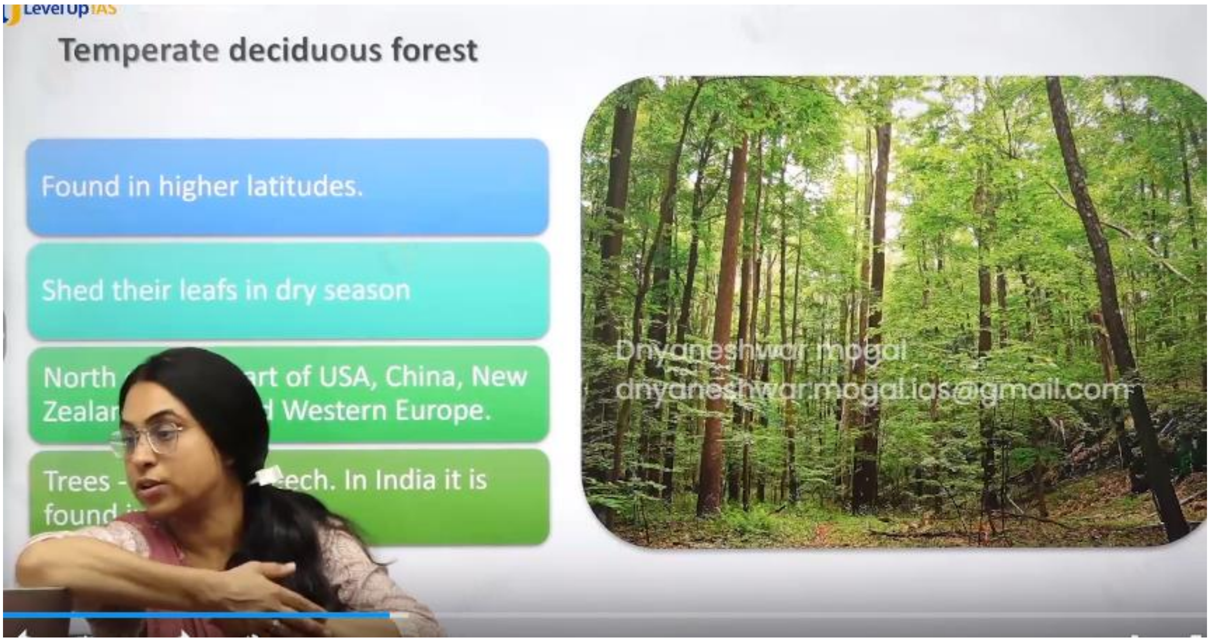
1. It is located in mid latitude and margin of continents. They comprise of sold wood and hard wood species but solfwood dominiate.
2. They are wild used for furniture and matchstick production paper production.
3. In India temperate every green forest are seen majorly in the Himalayas the name of species are oak pine eucalyptus.
4.



Tropical deciduous forest:

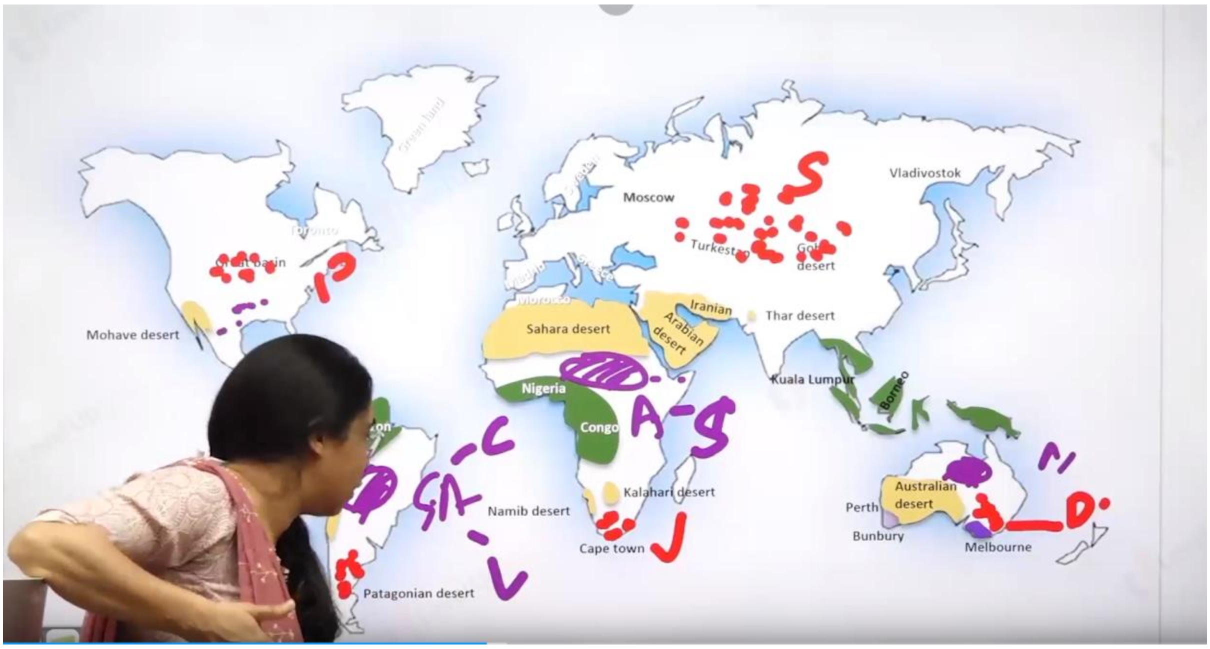
---

1. They are also known as monsoon forest
2. They are two types moist decedo00us forest and dry deciduous forest.
3. They are found in large part of India, Brazil, Mexico, some parts of Africa, Australia.
4. Tree shades their leave to conserve their moister. There is a mixture of softwood and hard wood. They are used for furniture.
5. Ex teak, sag, neem, Shisham.



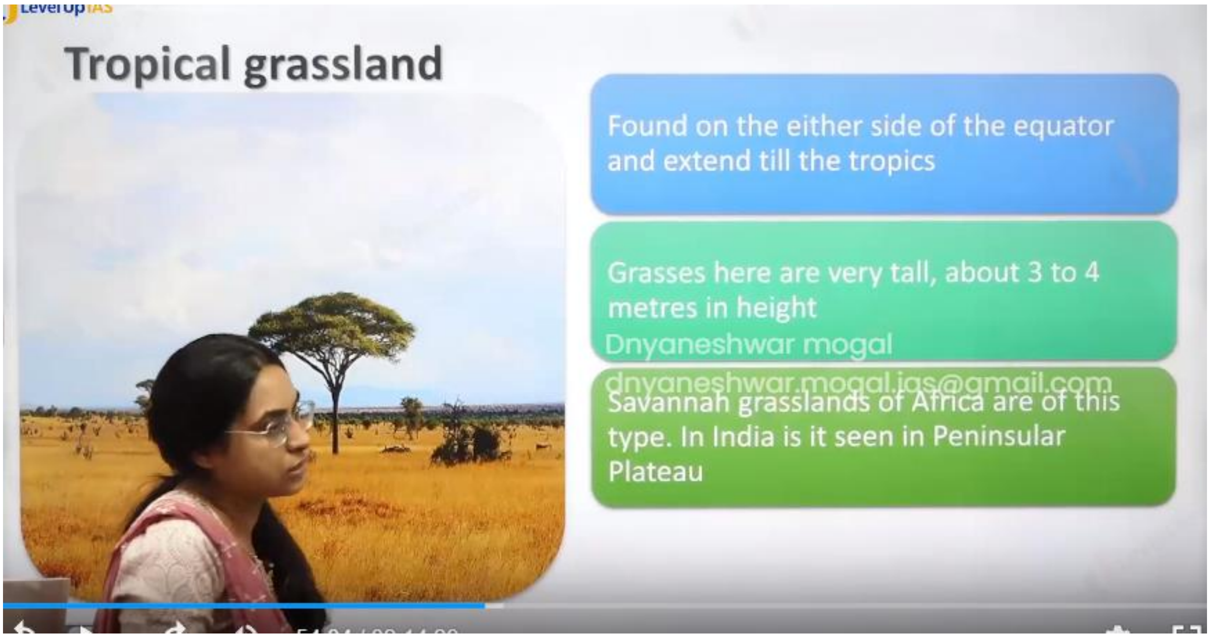
Temperate deciduous forest:

1. It is found in higher latitudes 15 degrees north and south.
2. Tree shade their leaves in dry season, They are found along the margin of temperate ever green forest, ex USA, China New Zeeland, western Europe, chili,
3. Species oak, ash, beech



The violet ones are tropical grassland

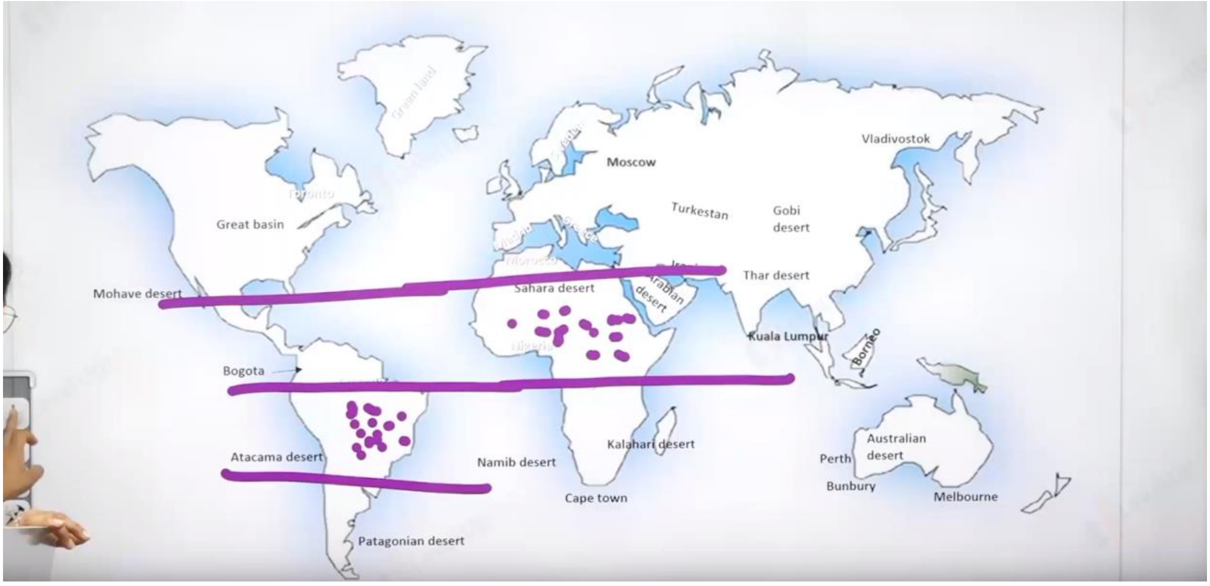
And the red ones are temperate grassland.

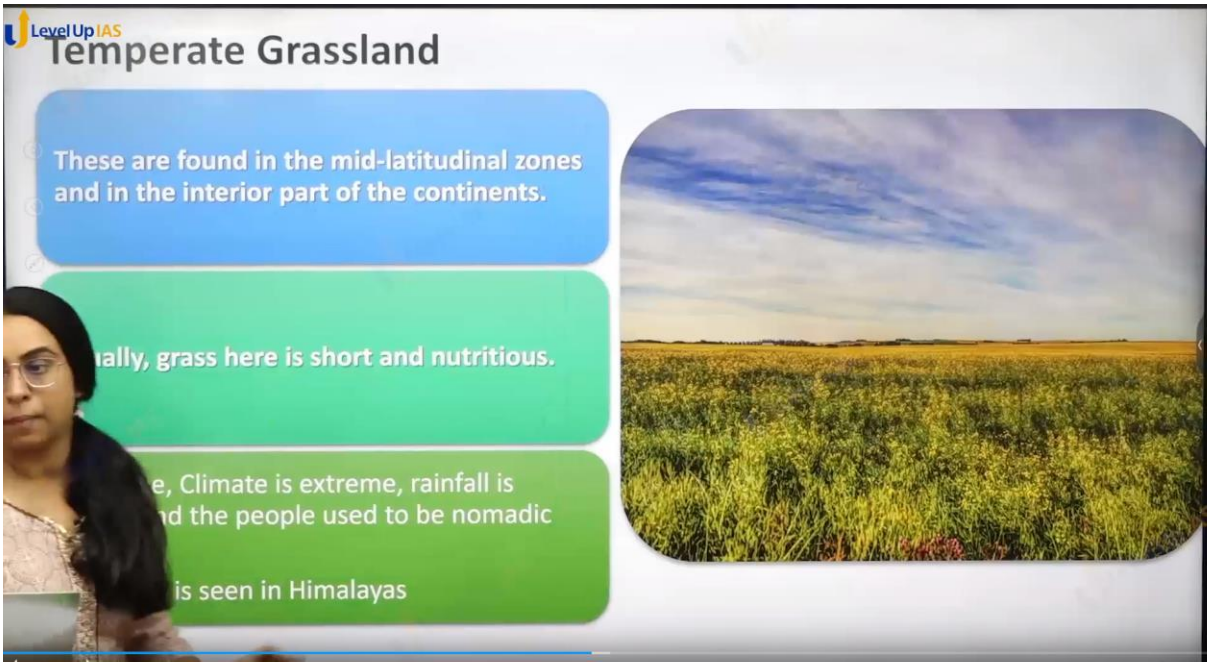


Tropical Grasslands:

---

1. They are found on either side of tropics, where there is moisture stress.
2. These grasslands are very tall and may reach 3 to 4 m height. Tropical grasslands are not pure grassland, there is a mixture of tree and grassland.
3. Grasses have become resistant to fire due to a lot of fire in the dry seasons, therefore this ecosystem is also called fire climax community.
4. Tropical grassland are not nutrients compared to temperate grasslands. Because grass need cold climate for better nutrition as well as the soil in region of tropical grassland Is not as nutrient rich
5. Ex: savannah in Africa, Llanos south America, Compos in south America.





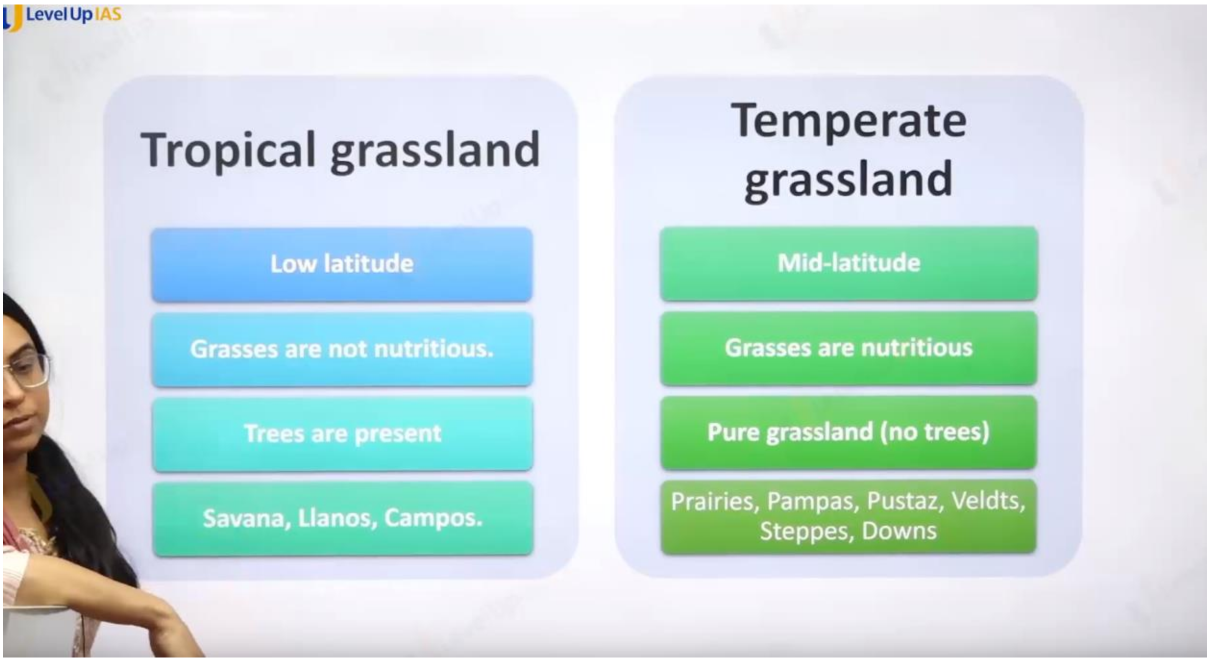
Temperate Grassland:

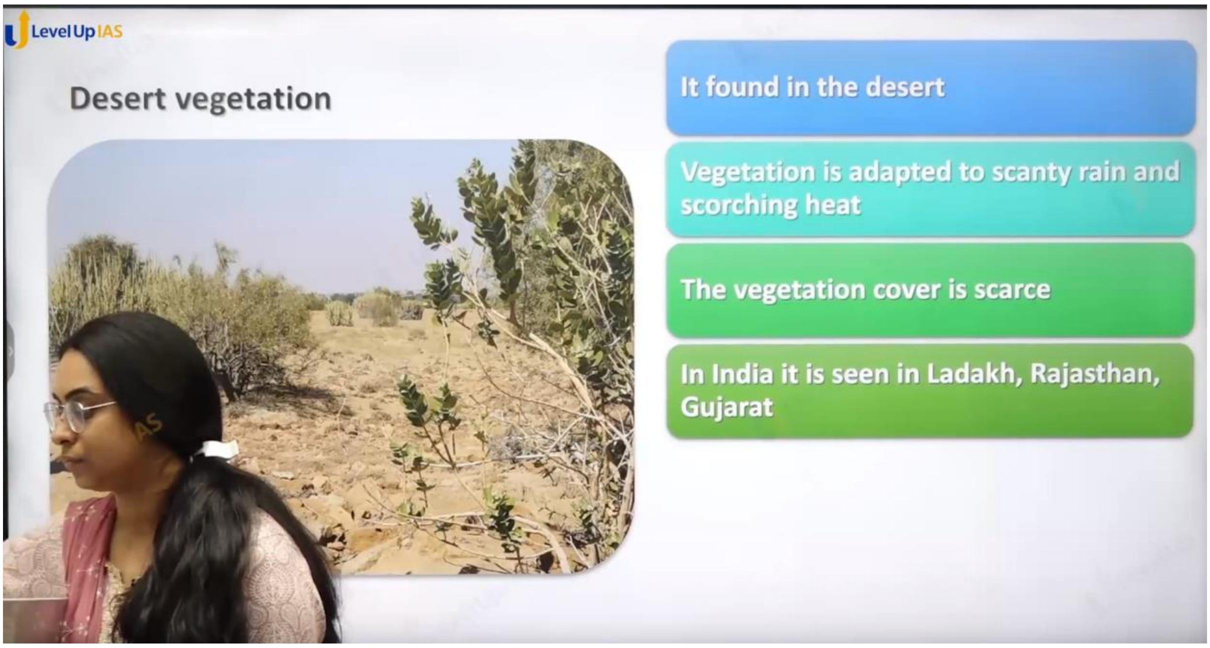
---

1. They are found in mid latitude zone in the inner part of continent
2. The grasses are usually short usually nutritious ex: steppes in Russia, pampas in Argentina, Pustaz in hungry, veldts in south Africa. Prairies

in north America (they are little on the wetter side) and downs in Austria.
3. They are pure grassland unlike savannah.
4.

Difference

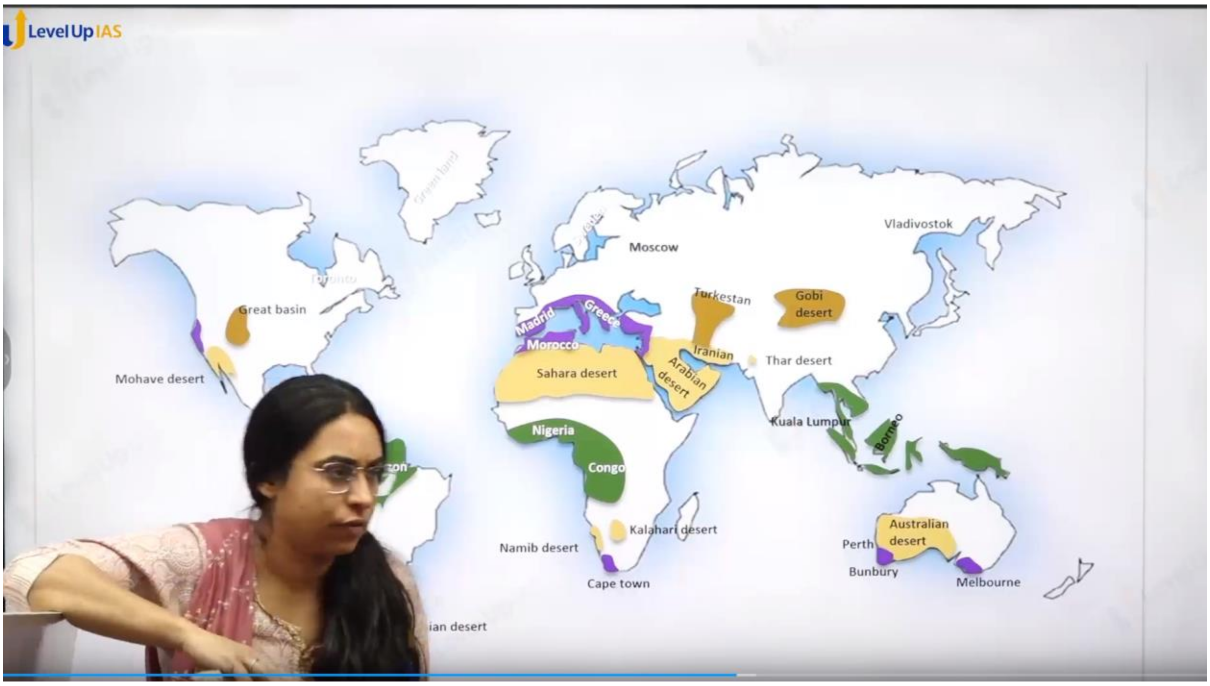




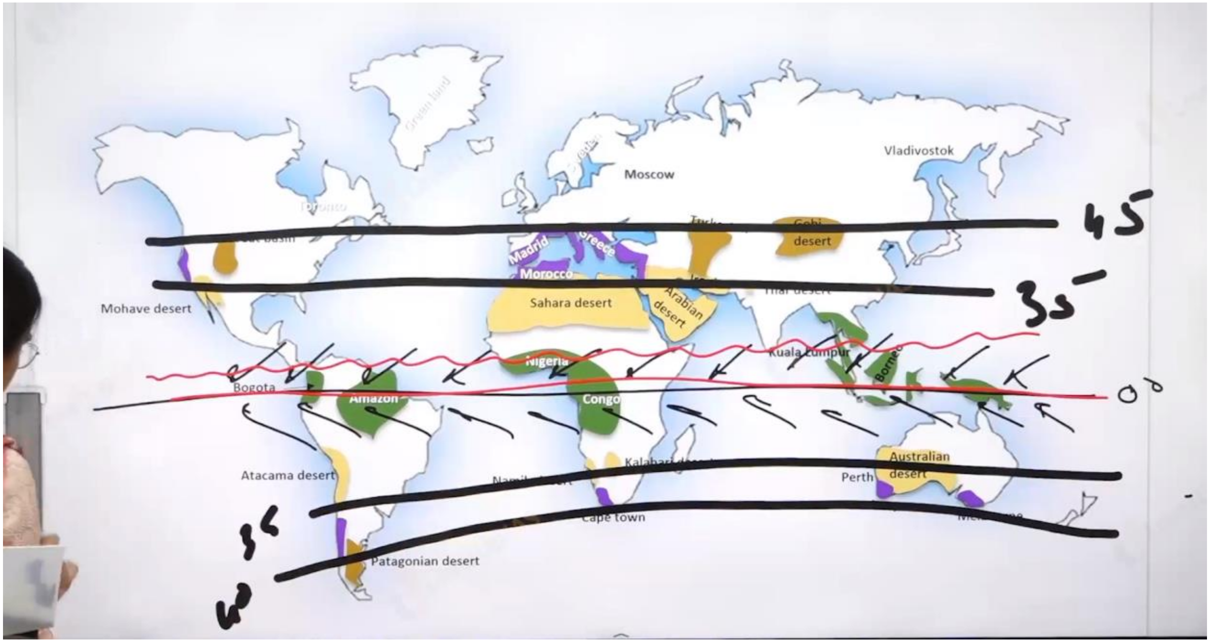
Desert Vegetation:

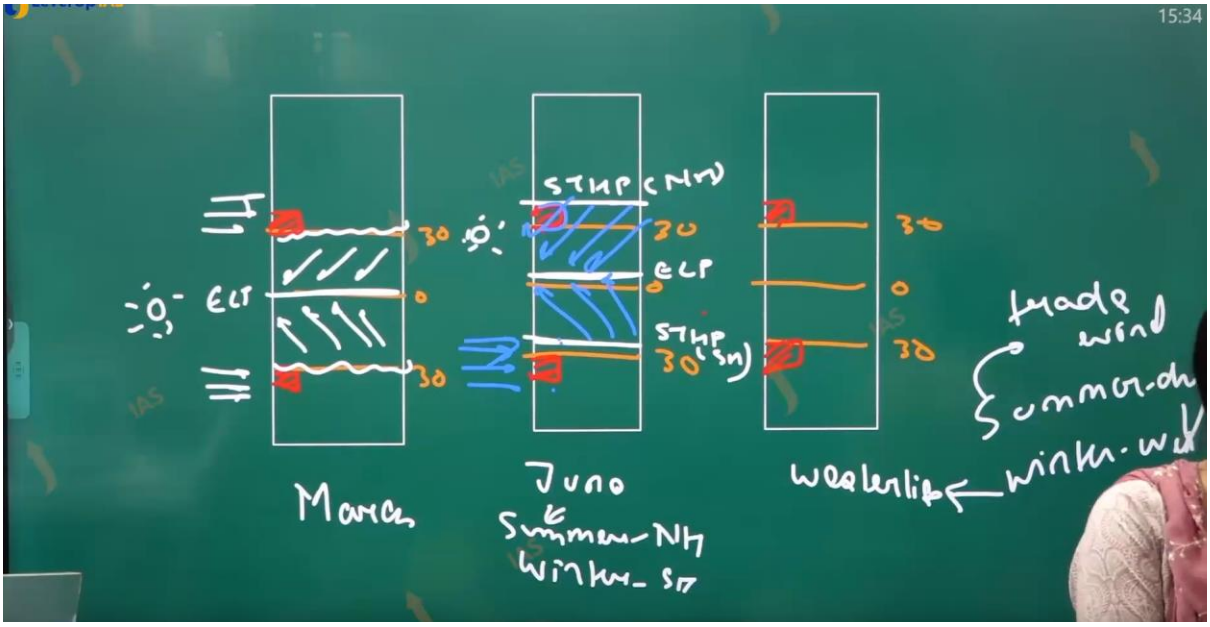
---

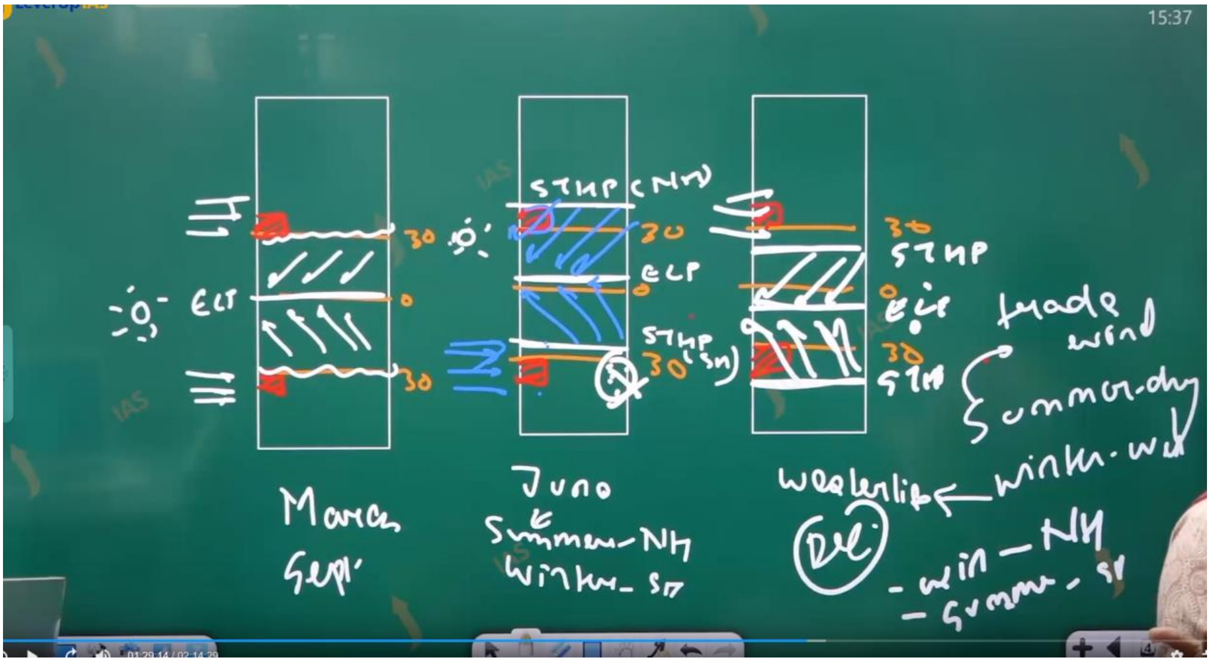
1. It was found in the desert. They vegetation scars and vegetation are adapted to water stress. In India desert vegetation is seen In the Ladakh, Rajasthan, Gujarat.
2. The tree adaptation is like long cap roots to reach ground water, horizontal to search water. Seed lies dormmate for years, and quickly germinate when it rains. Tree store water in the root stem leaves fruit and they are called succulent.
3. Leaves they become small and they can become Thorn or they can become waxy and leather with sunken stomata to conserver the water loss.

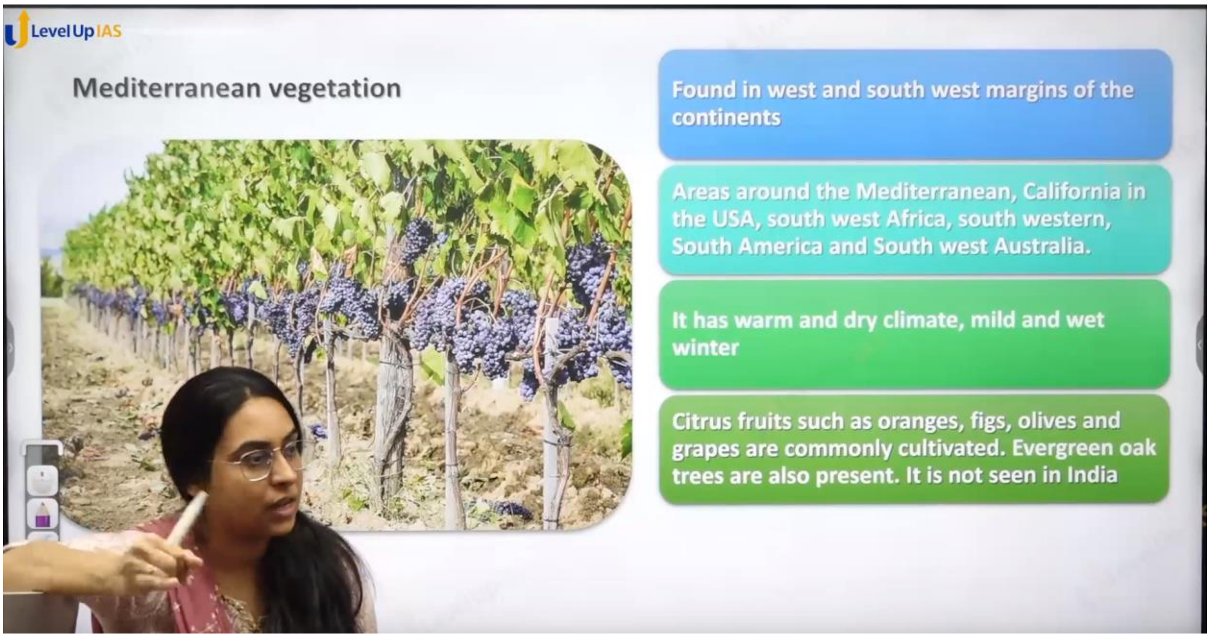


It is show in the purple color.




15:34
STHP (NH)
30
ELP
0
STHP (SH)
30
ELT
30
0
30
March
June
Summer - NH
Winter - SH
trade wind
Summer - dry
Winter - wet
Westerlies

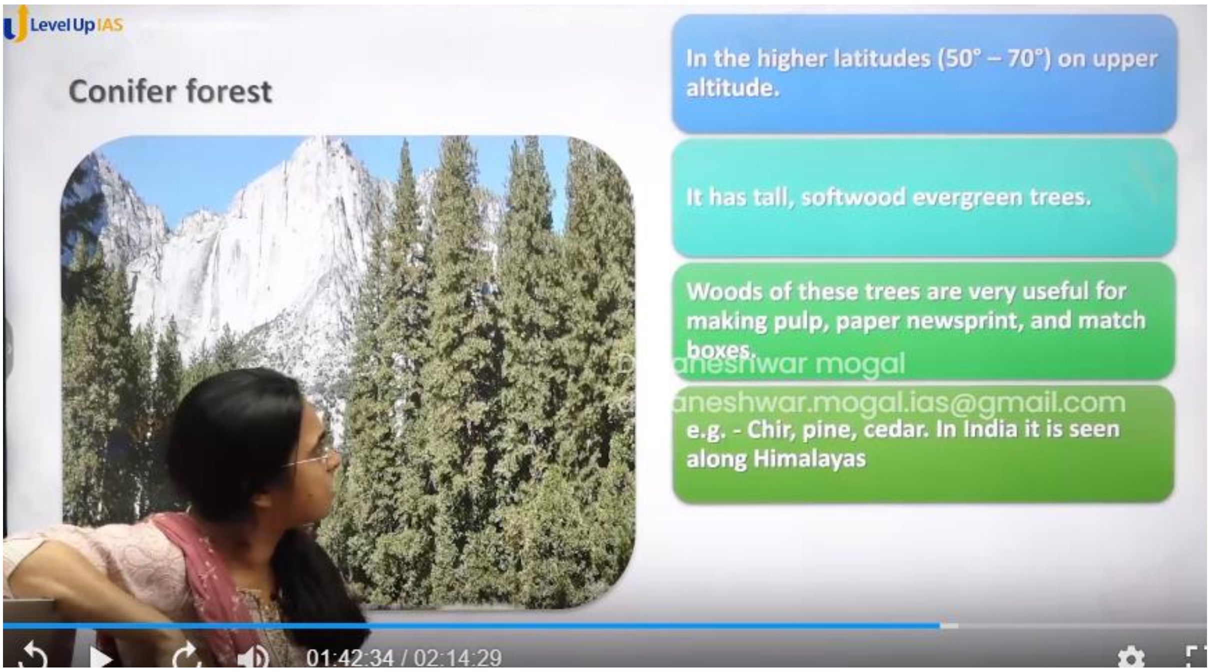

15:37
STHP (NH)
30
ELP
0
STHP (SH)
30
ELT
30
STHP
0
ELP
30
STH
March
Sept
June
Summer - NH
Winter - SH
trade wind
Summer - dry
Winter - wet
Westerlies
- win - NH
- summer - SH



Mediterranean vegetation:

---

1. It found the in the western side of continent between 35 – 43 degrees north and south.
2. It was found in the Mediterranean climate which has special conditions like rainfall in winter due to westerlies winds dry condition in the summer due to offshore trade winds.
3. The Change in the pattern is due to shifting pressure belt with apparent shift of sun.
4. It seen the area of Mediterranean coast, California of USA, southwest Africa, southwest Africa, southwest America, the southern part of Australia.
5. The vegetation grown here are citrus fruits like oranges and gigs olive jalapeno, grapes, ever green oak tree also present this type of vegetation is not seen in India.
6. It also known as sclerophylls



Conifer forest/ vegetation:

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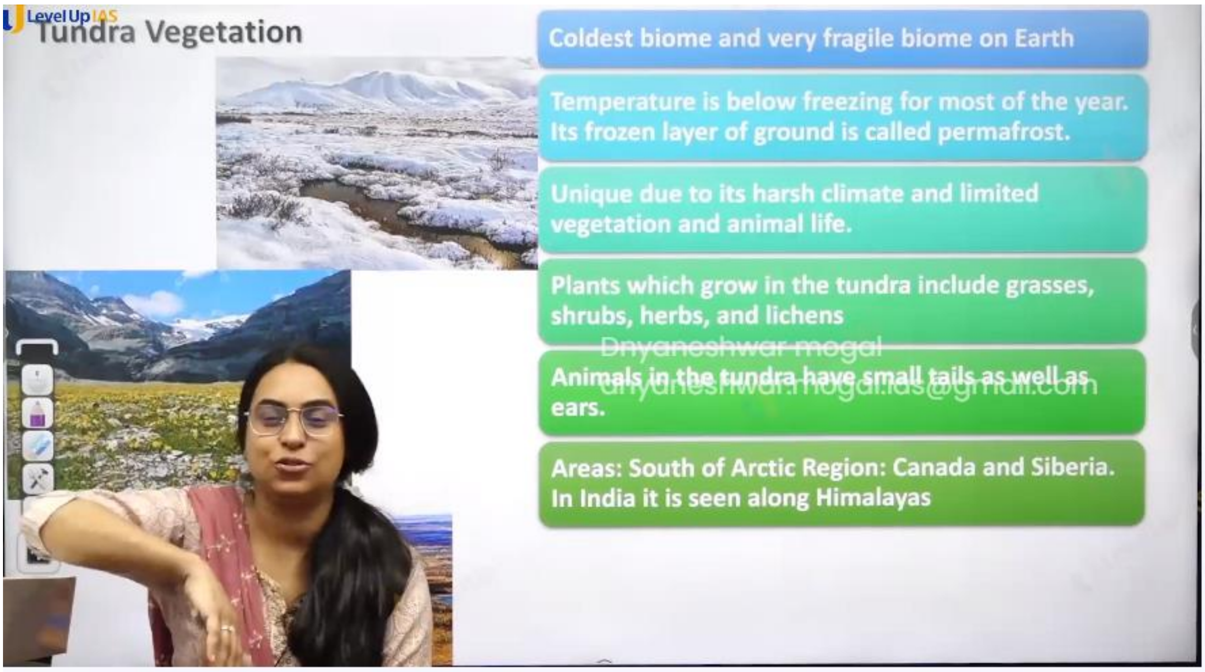
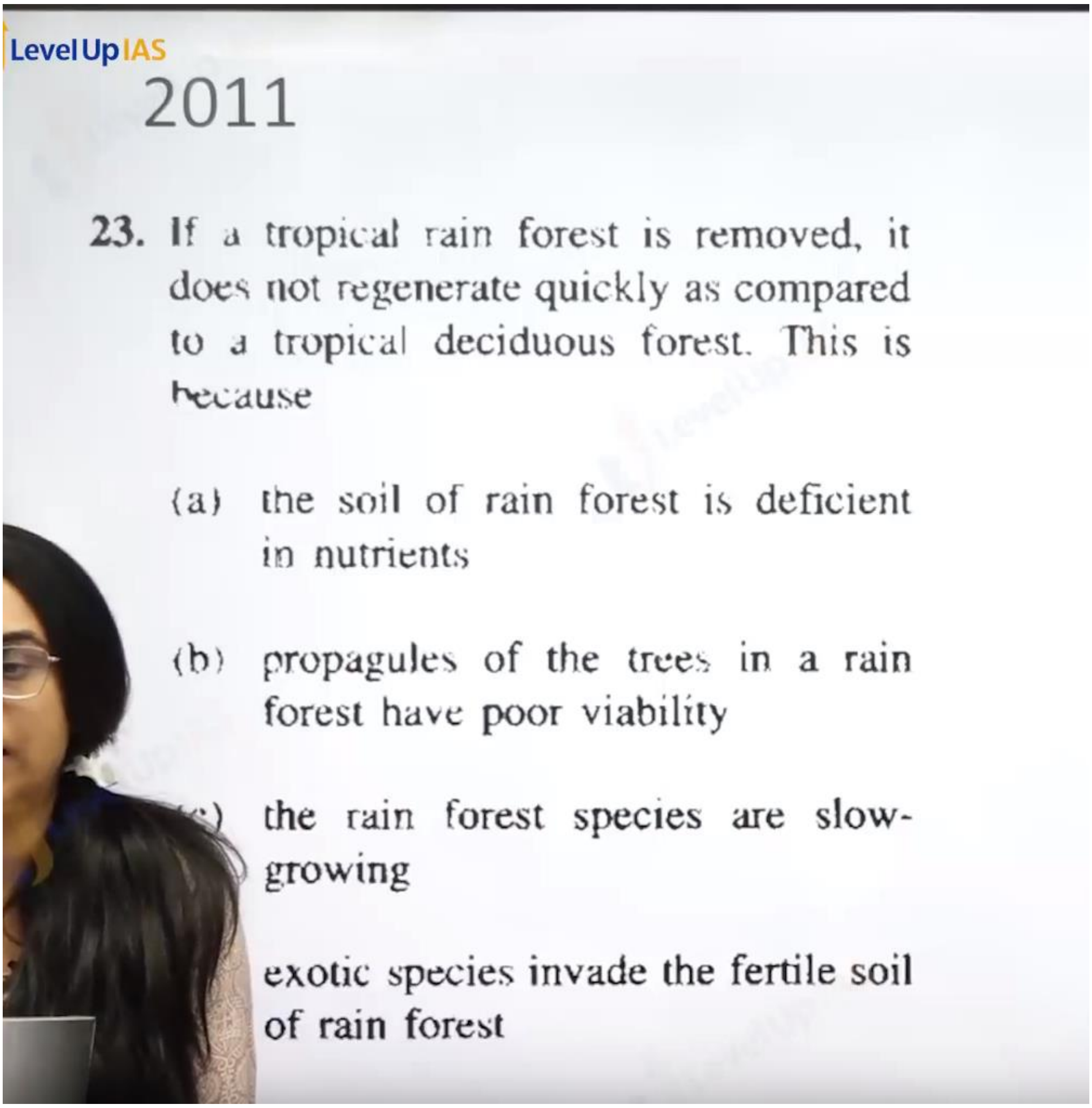
1. It was found in the higher latitudes 60-70 degrees north and in upper latitudes. It has tall soft wood ever green trees. Wood is used for making pulp paper newspapers. Chir pine and cedar it seen along the Himalayas.



Tundra Vegetation:

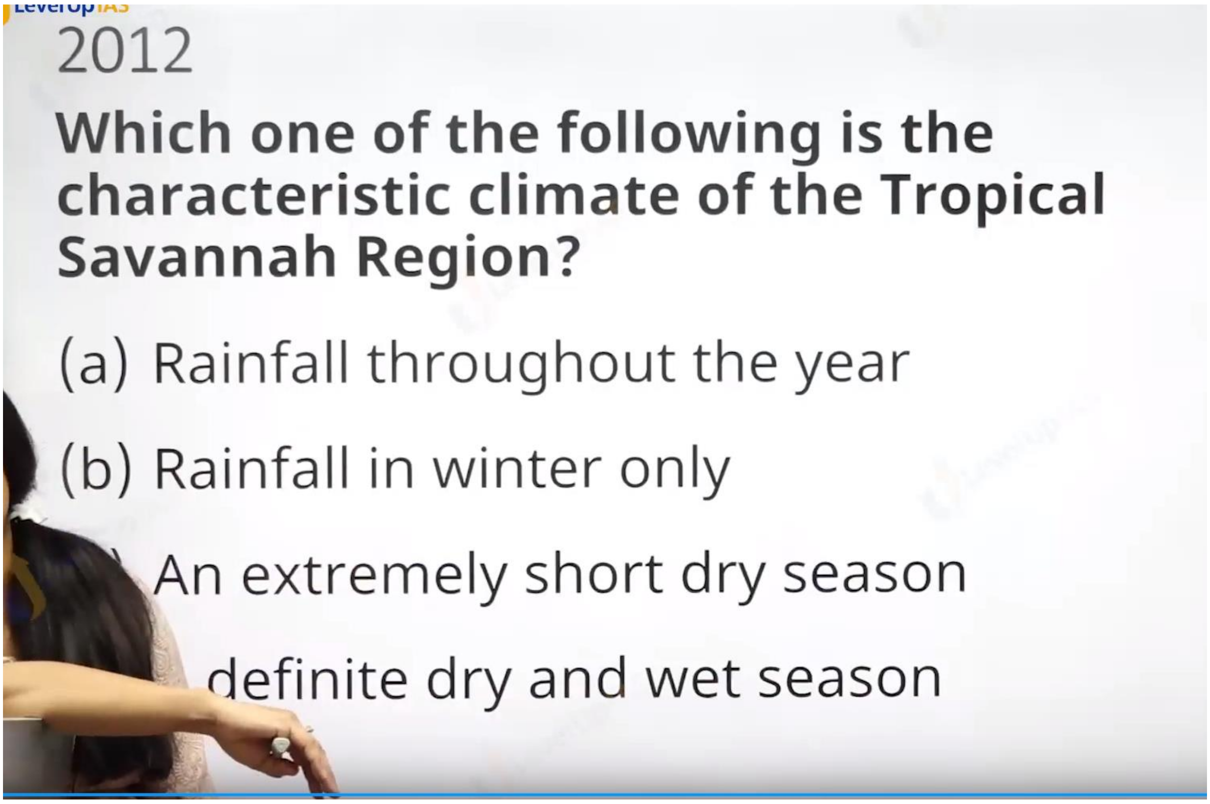
---

1. Coldest biome. And very fragile biome on earth
2. The temperature is below the freezing point of water for most of the year and therefore the frozen layer of ground is called frame forest.
3. It is a unique vegetation which adapted to harsh life /climate therefore it has limited vegetation and animal life.
4. The plants that grow in tundra vegetation are grass herbs lichen shrubs.
5. The animal in tundra is small tail and small ear. It found in the region in the Canada Siberia greenland Himalayas.

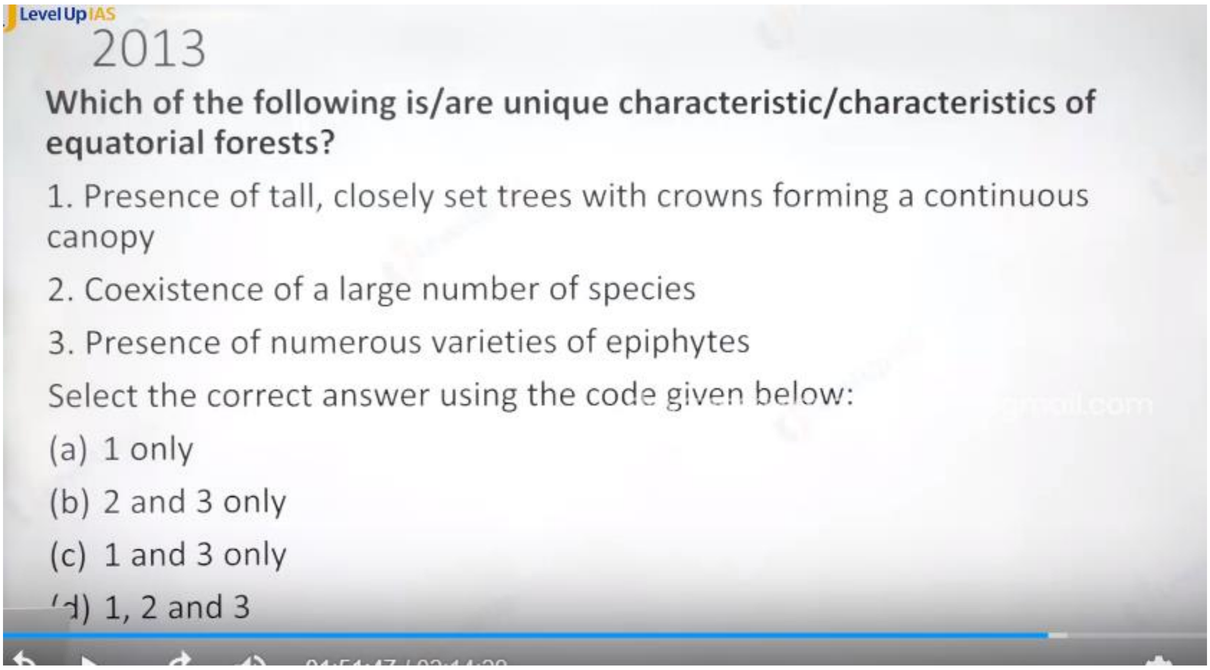


Ans A.

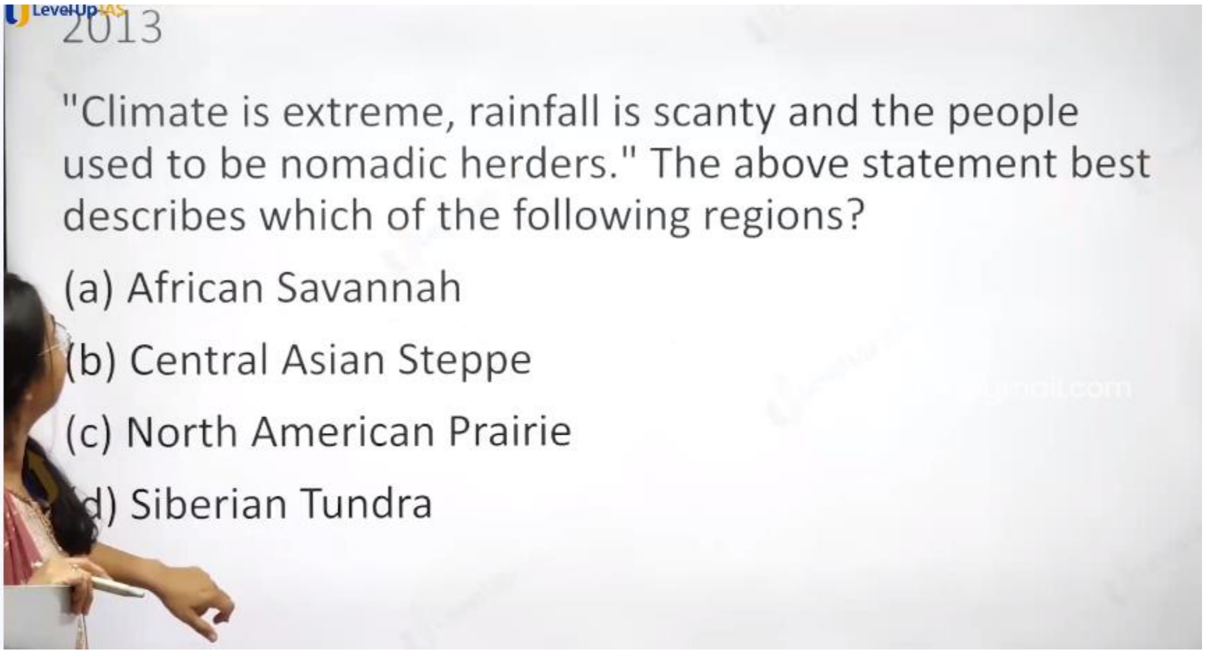
Vegetation of India:


2012
Which one of the following is the characteristic climate of the Tropical Savannah Region?
(a) Rainfall throughout the year
(b) Rainfall in winter only
An extremely short dry season
definite dry and wet season

Ans d


Level Up IAS
2013
Which of the following is/are unique characteristic/characteristics of equatorial forests?
1. Presence of tall, closely set trees with crowns forming a continuous canopy
2. Coexistence of a large number of species
3. Presence of numerous varieties of epiphytes
Select the correct answer using the code given below:
(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
d) 1, 2 and 3

Ans Is D

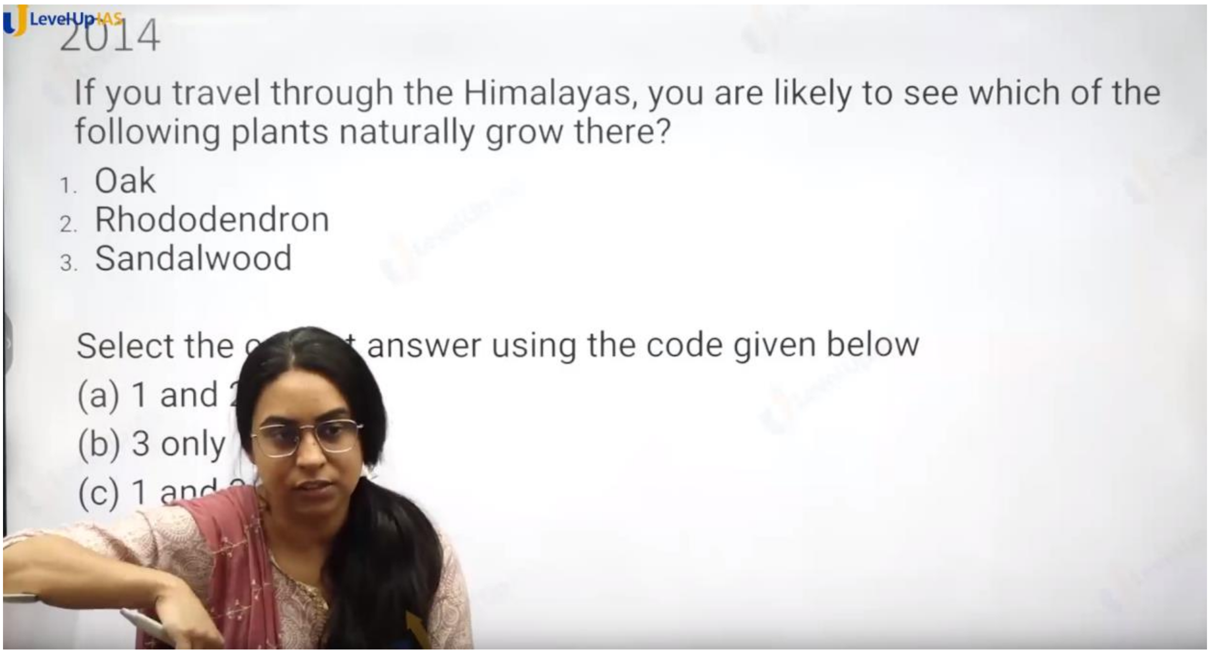


Ans B.

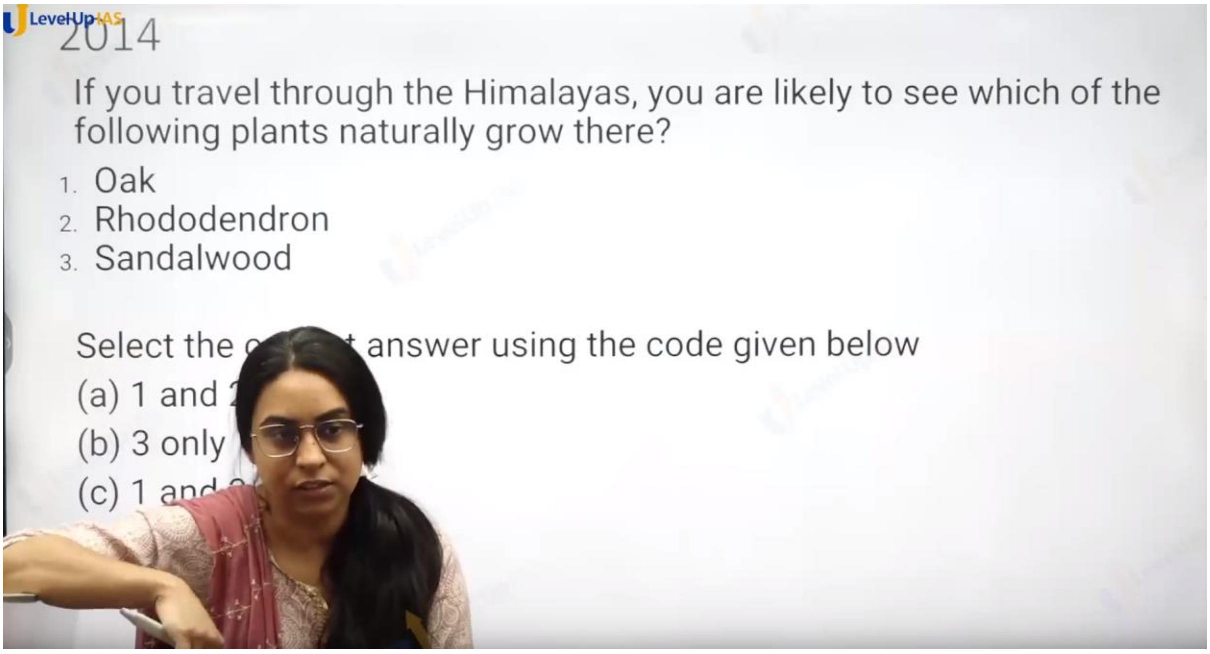
Trundra do not have the livestocks. Not d

Prairie is good rainfall.

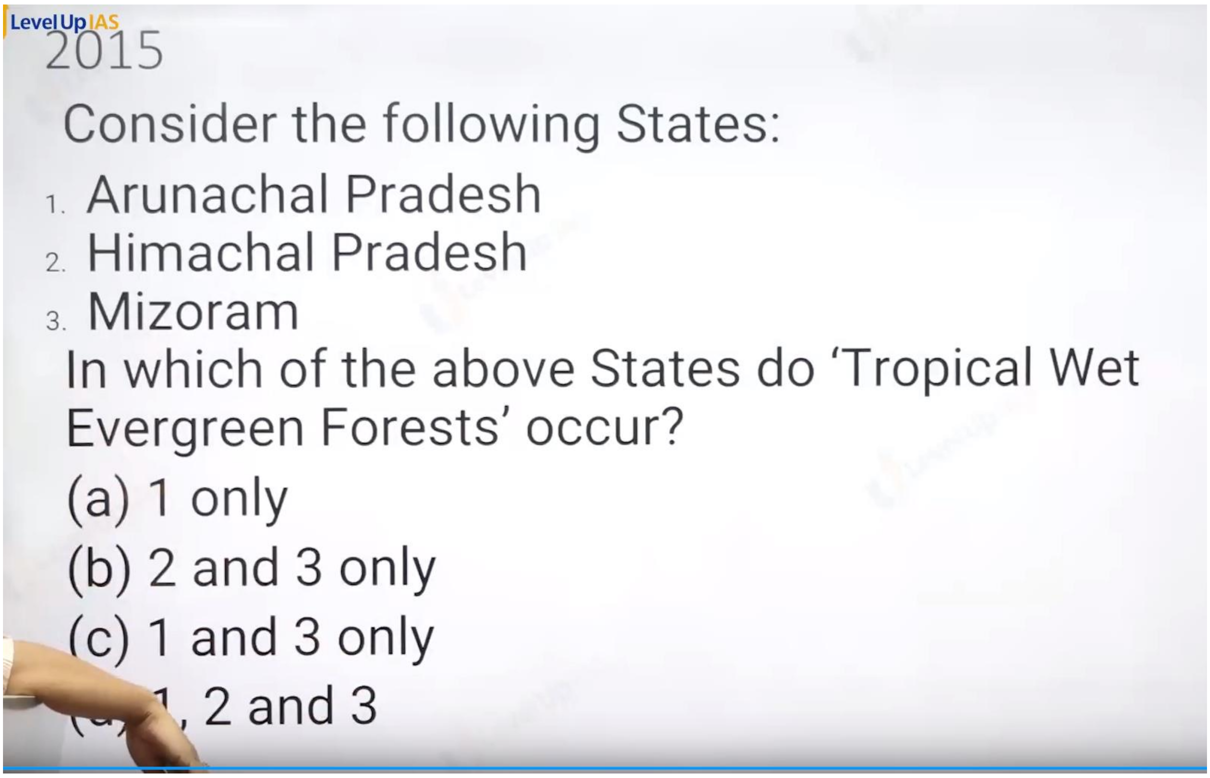
Savanna not extreme climate.



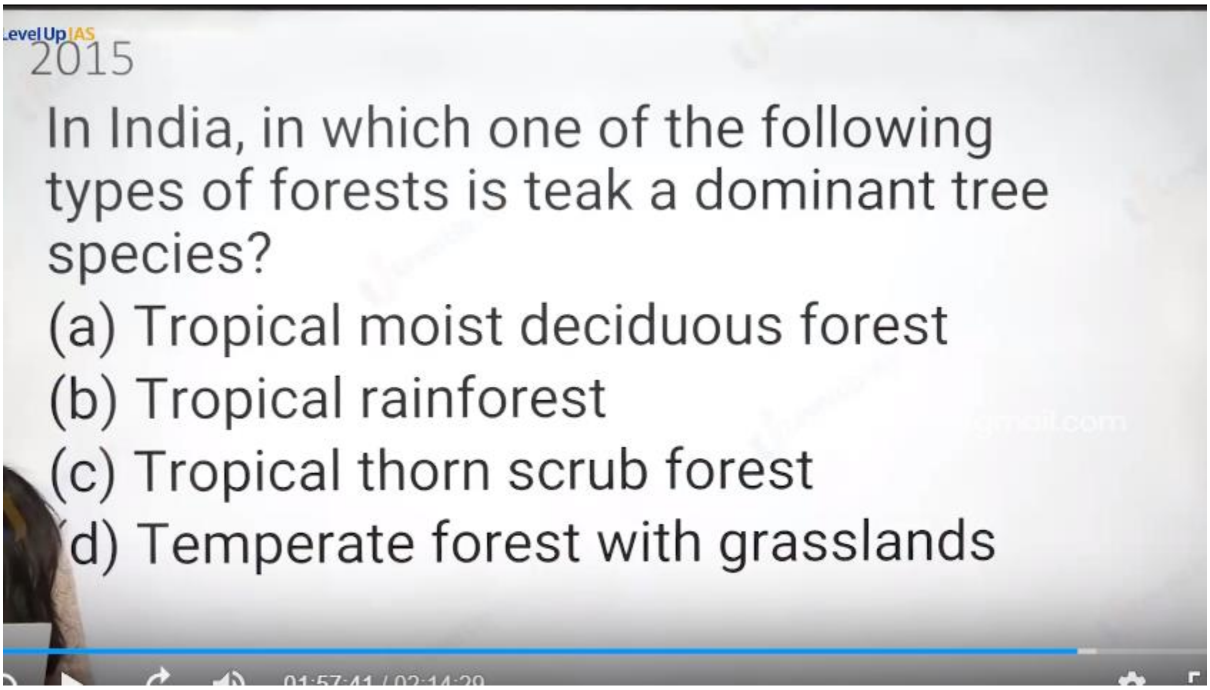
Ans is a not sandhood.



Meghalaya.



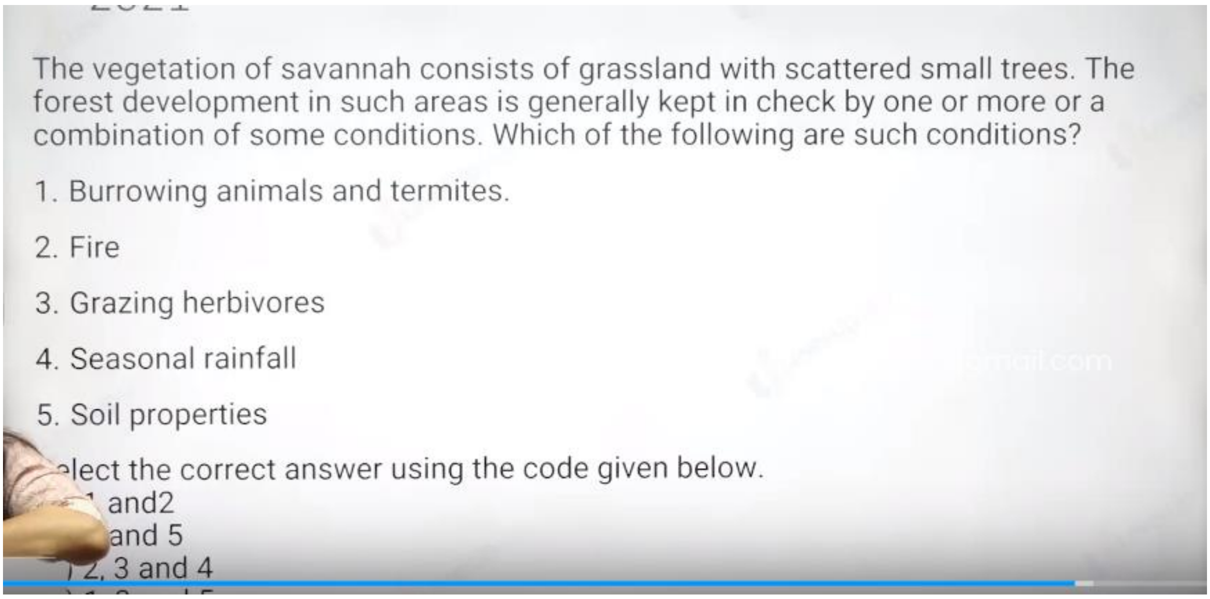
Ans is C.



Ans a.

Tropical evergreen forest : rosewood mahagani ebony

Tropical throono : babool cactus



Ans id C

Level Up IAS

2021

Leaf litter decomposes faster than in any other biome and as a result, the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees." This is the most likely description of

(a) Coniferous forest
(b) Dry deciduous forest
(c) Mangrove forest
(d) Tropical rain forest

Level Up IAS

2023

Consider the following statements:

Statement-I: The soil in tropical rain forests is rich in nutrients.

Statement-II: The high temperature and moisture of tropical rainforests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the abo ents?

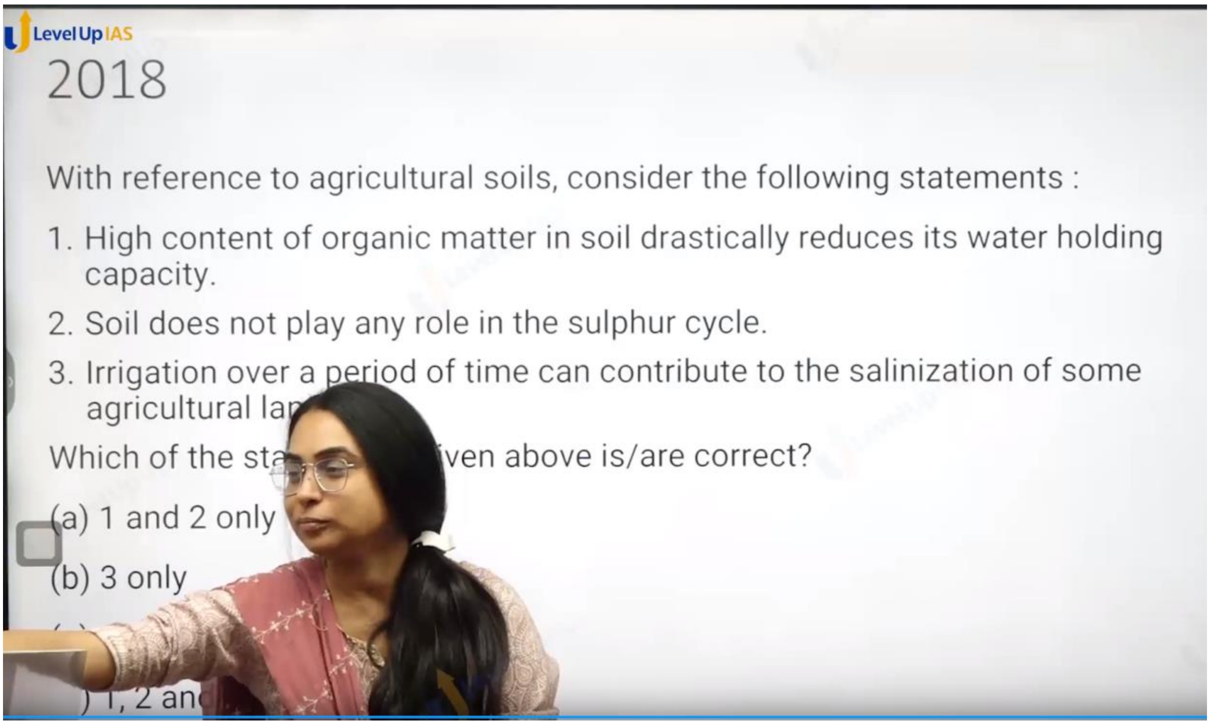
a) Both Statement-I and Statement-II are correct and is Correct the explanation for Statement-I

b) Both Statement-I and Statement-II are correc not the correct explanation for Statement-I

Statement-I is correct but Statement-II is incor

02:05:14 / 02:14:29

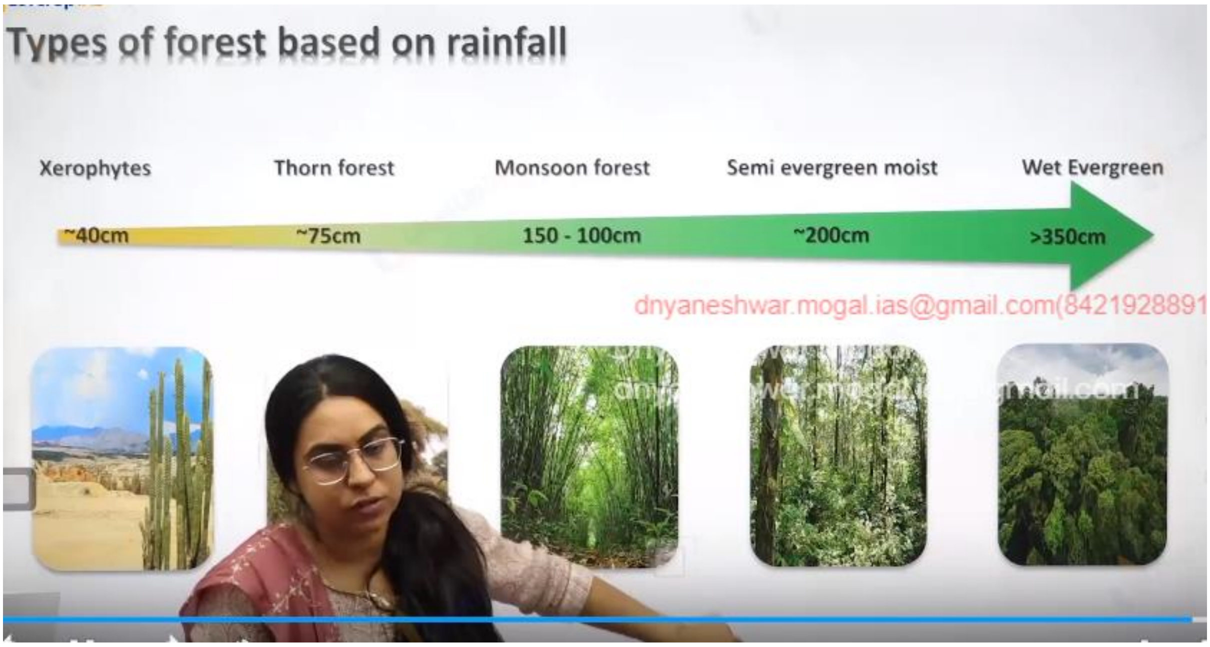
Ans d. first is wrong



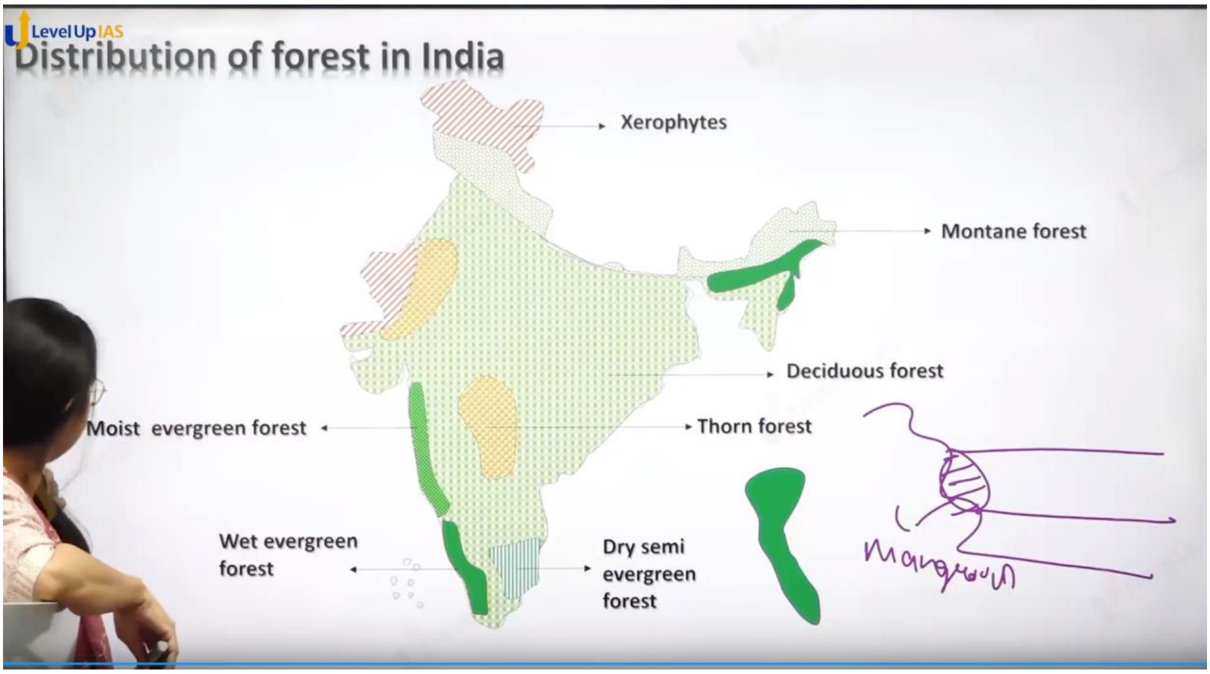
Vegetation of India:

---

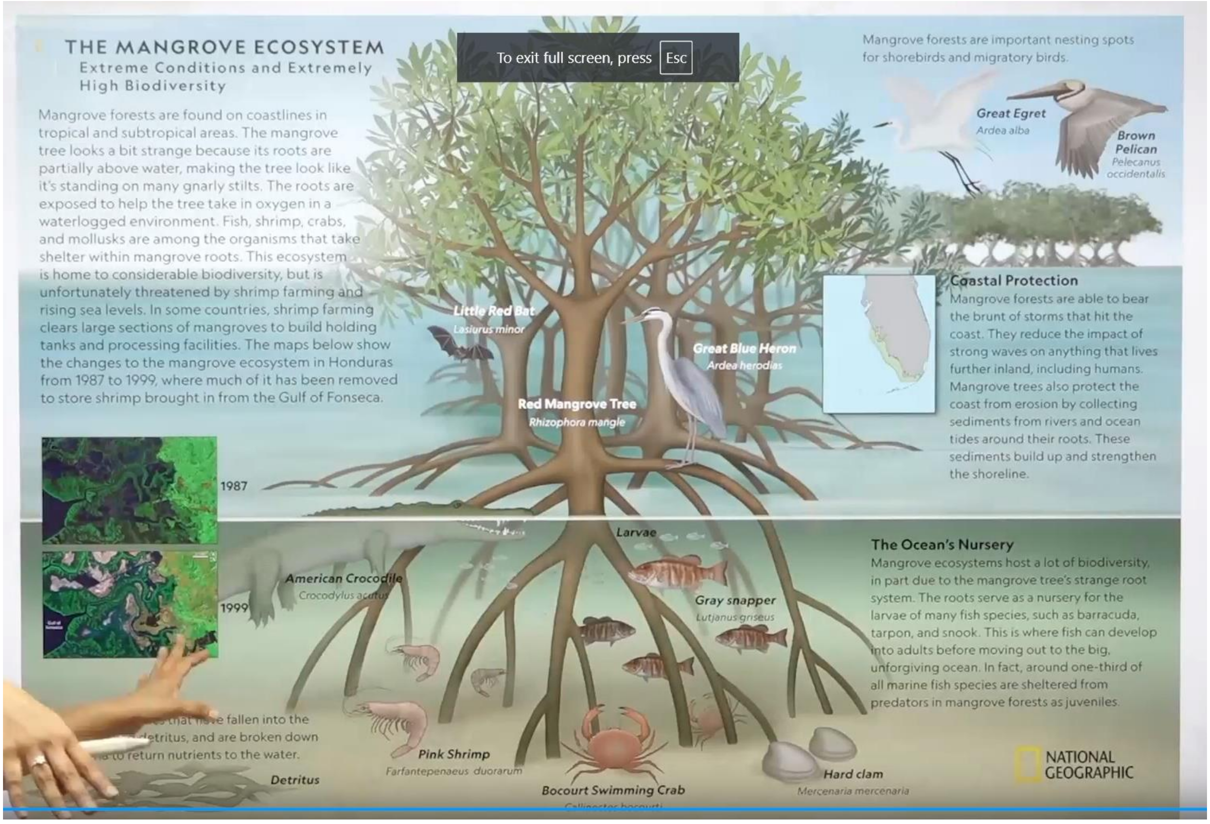
1. You have wet ever green forest, moist ever green forest. Monsoon forest, deciduous forest, throne forest. xerophyte. Montaine forest, mangroves: -shola forest.
2. Wet ever green forest: rainfall > 350 cm/ Year
3. Moist ever green forest: rainfall >250 cm /year
4. Dry semi ever green forest: 200 cm
5. Monsoon forest: rainfall 75 -150 cm/year.
6. Dry deciduous forest: 75-100
7. moist deciduous forest: 100- 150
8. thorn forest: 75
9. xerophyte: 45 :
10. mangroves are : tidal halo phytic plant.

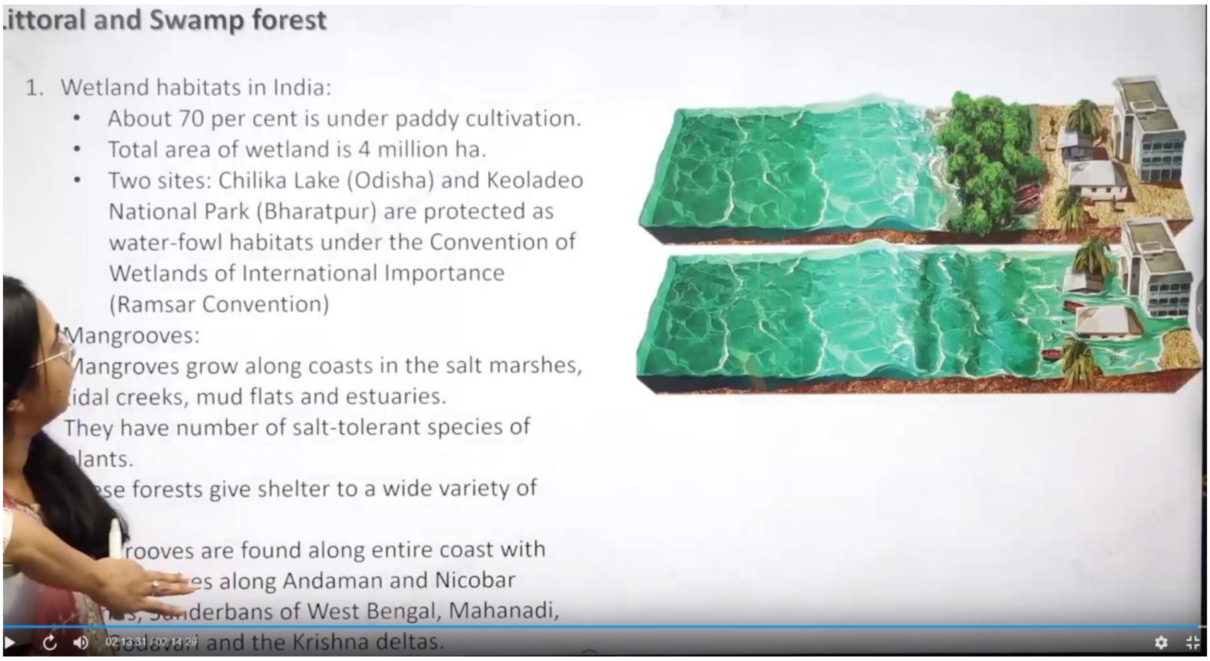


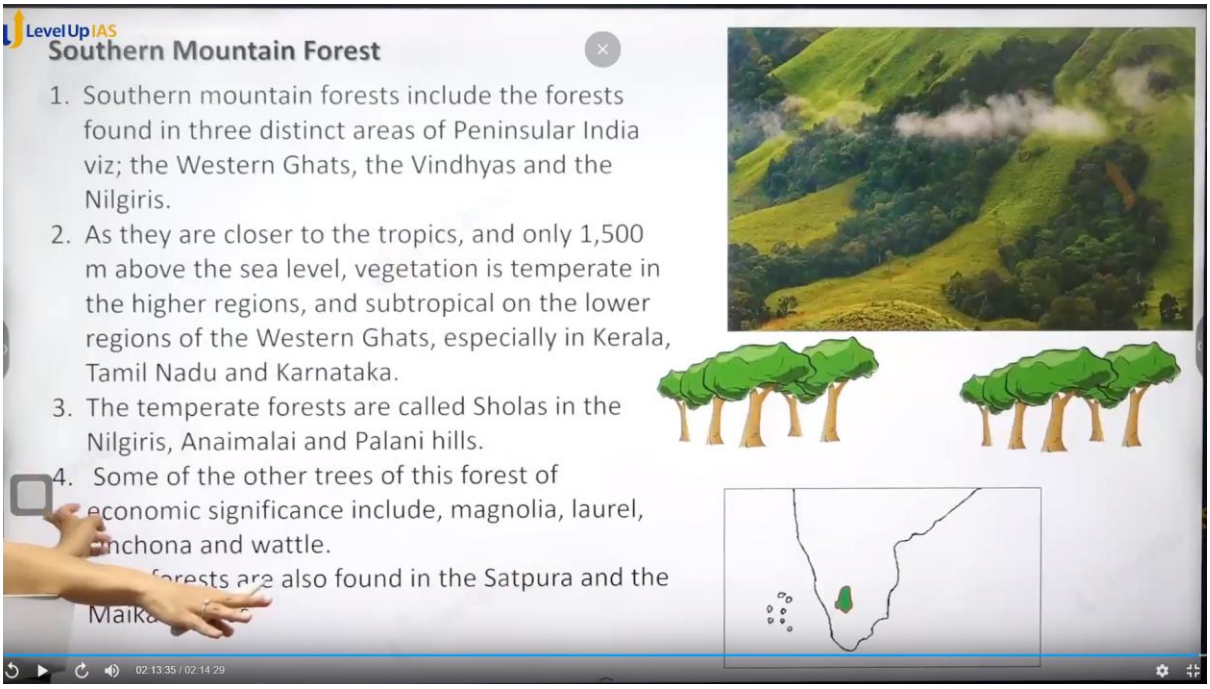
Types of forest based on rainfall
Xerophytes
Thorn forest
Monsoon forest
Semi evergreen moist
Wet Evergreen
~40cm
~75cm
150 - 100cm
~200cm
>350cm
dnyaneshwar.mogal.ias@gmail.com(8421928891



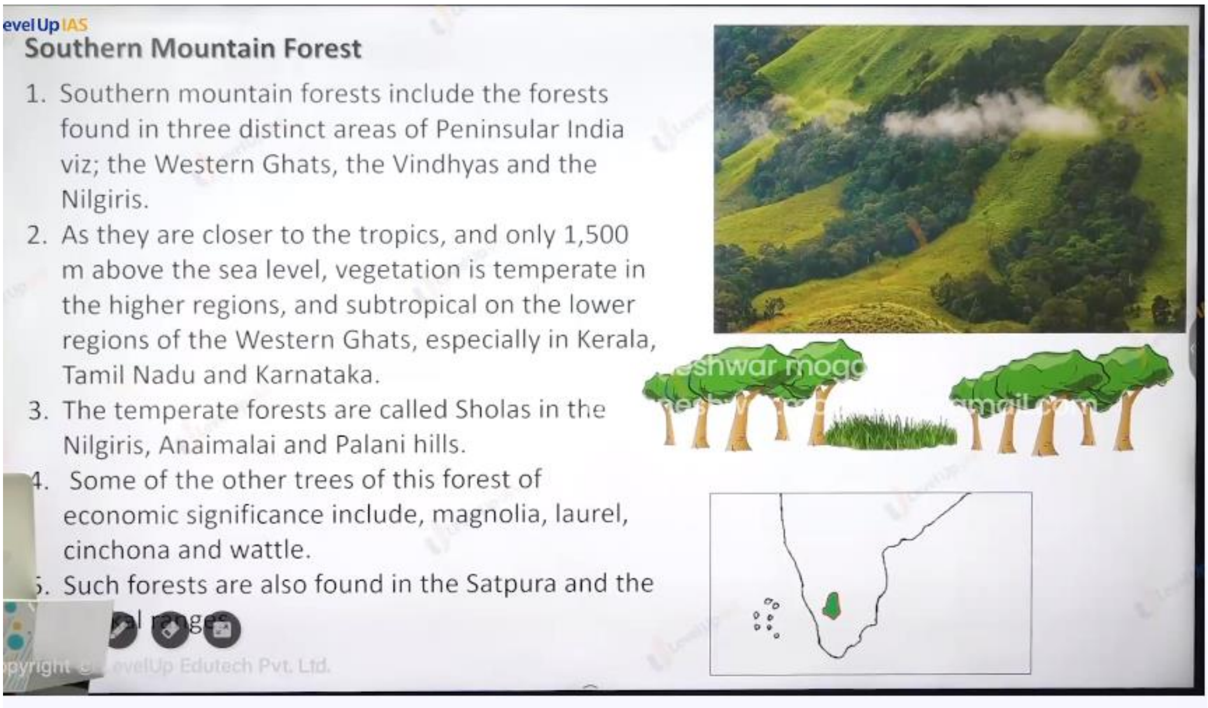
Level Up IAS
Distribution of forest in India
Xerophytes
Montane forest
Deciduous forest
Moist evergreen forest
Thorn forest
Wet evergreen forest
Dry semi evergreen forest
Mangroov


To exit full screen, press Esc
THE MANGROVE ECOSYSTEM
Extreme Conditions and Extremely High Biodiversity
Mangrove forests are found on coastlines in tropical and subtropical areas. The mangrove tree looks a bit strange because its roots are partially above water, making the tree look like it's standing on many gnarly stilts. The roots are exposed to help the tree take in oxygen in a waterlogged environment. Fish, shrimp, crabs, and mollusks are among the organisms that take shelter within mangrove roots. This ecosystem is home to considerable biodiversity, but is unfortunately threatened by shrimp farming and rising sea levels. In some countries, shrimp farming clears large sections of mangroves to build holding tanks and processing facilities. The maps below show the changes to the mangrove ecosystem in Honduras from 1987 to 1999, where much of it has been removed to store shrimp brought in from the Gulf of Fonseca.
1987
1999
Mangrove forests are important nesting spots for shorebirds and migratory birds.
Great Egret
Ardea alba
Brown Pelican
Pelecanus occidentalis
Little Red Bat
Lasiurus minor
Great Blue Heron
Ardea herodias
Red Mangrove Tree
Rhizophora mangle
Coastal Protection
Mangrove forests are able to bear the brunt of storms that hit the coast. They reduce the impact of strong waves on anything that lives further inland, including humans. Mangrove trees also protect the coast from erosion by collecting sediments from rivers and ocean tides around their roots. These sediments build up and strengthen the shoreline.
Larvae
American Crocodile
Crocodylus acutus
Gray snapper
Lutjanus griseus
The Ocean's Nursery
Mangrove ecosystems host a lot of biodiversity, in part due to the mangrove tree's strange root system. The roots serve as a nursery for the larvae of many fish species, such as barracuda, tarpon, and snook. This is where fish can develop into adults before moving out to the big, unforgiving ocean. In fact, around one-third of all marine fish species are sheltered from predators in mangrove forests as juveniles.
Detritus
Pink Shrimp
Farfantepenaeus duorarum
Bocourt Swimming Crab
Hard clam
Mercenaria mercenaria
NATIONAL GEOGRAPHIC


Littoral and Swamp forest
1. Wetland habitats in India:
• About 70 per cent is under paddy cultivation.
• Total area of wetland is 4 million ha.
• Two sites: Chilika Lake (Odisha) and Keoladeo National Park (Bharatpur) are protected as water-fowl habitats under the Convention of Wetlands of International Importance (Ramsar Convention)
Mangrooves:
Mangroves grow along coasts in the salt marshes, idal creeks, mud flats and estuaries.
They have number of salt-tolerant species of lants.
se forests give shelter to a wide variety of
rooves are found along entire coast with
es along Andaman and Nicobar
nderbans of West Bengal, Mahanadi,
and the Krishna deltas.

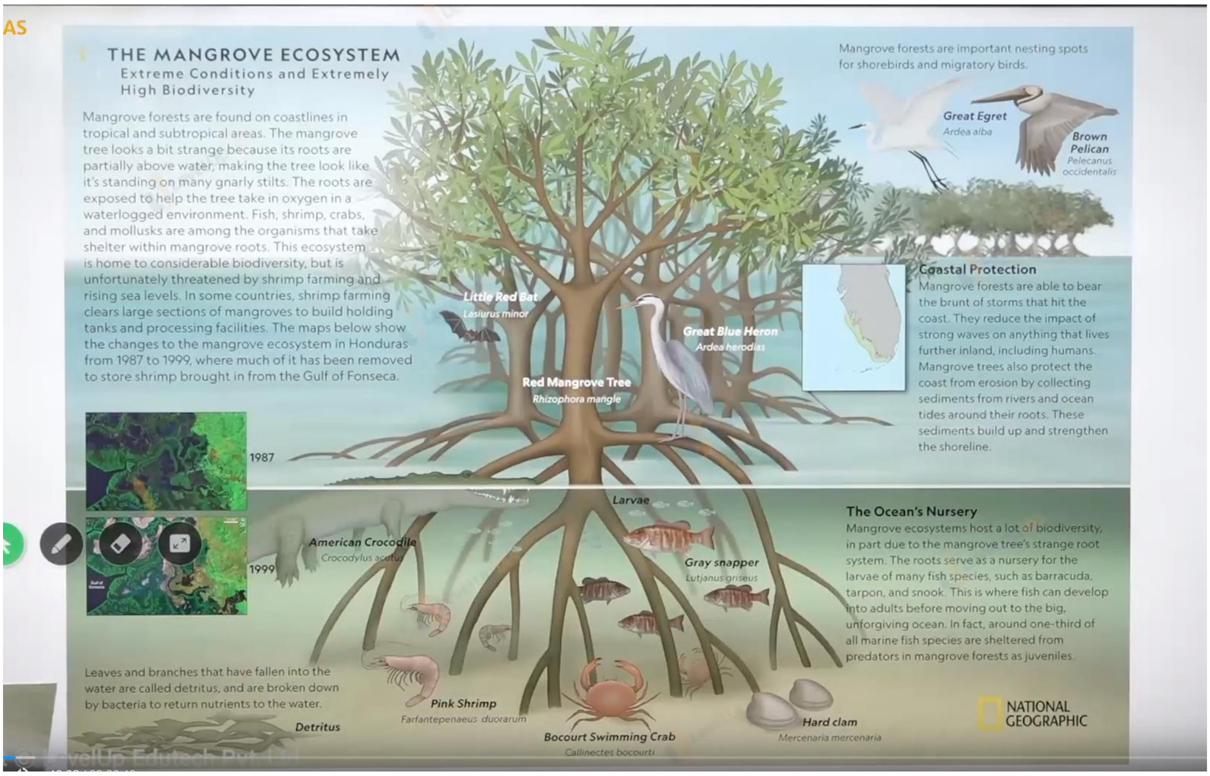


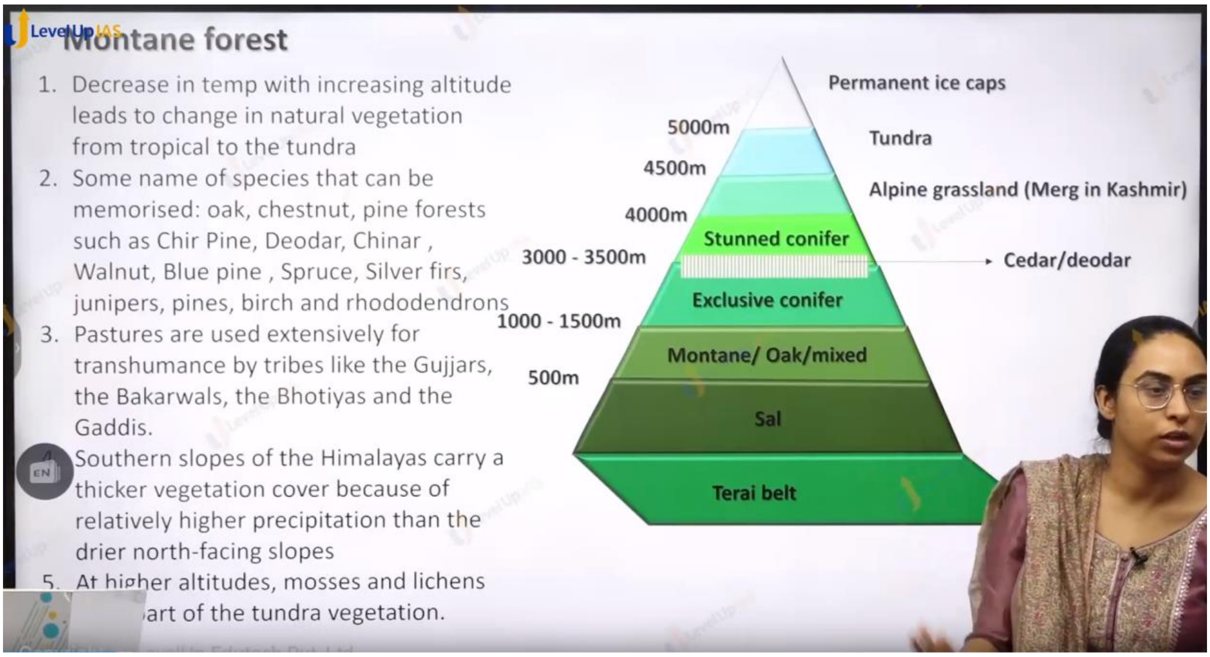
(35) Vegetation and Soil-II



Southern Mountain Forest:

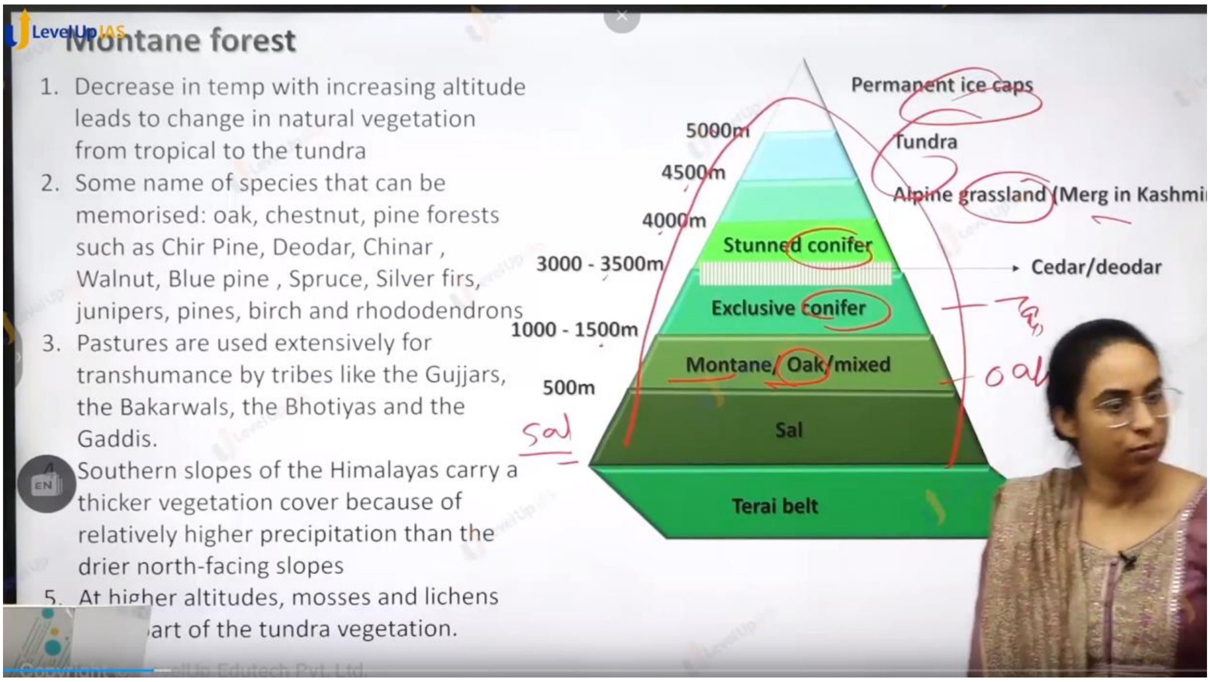
1.

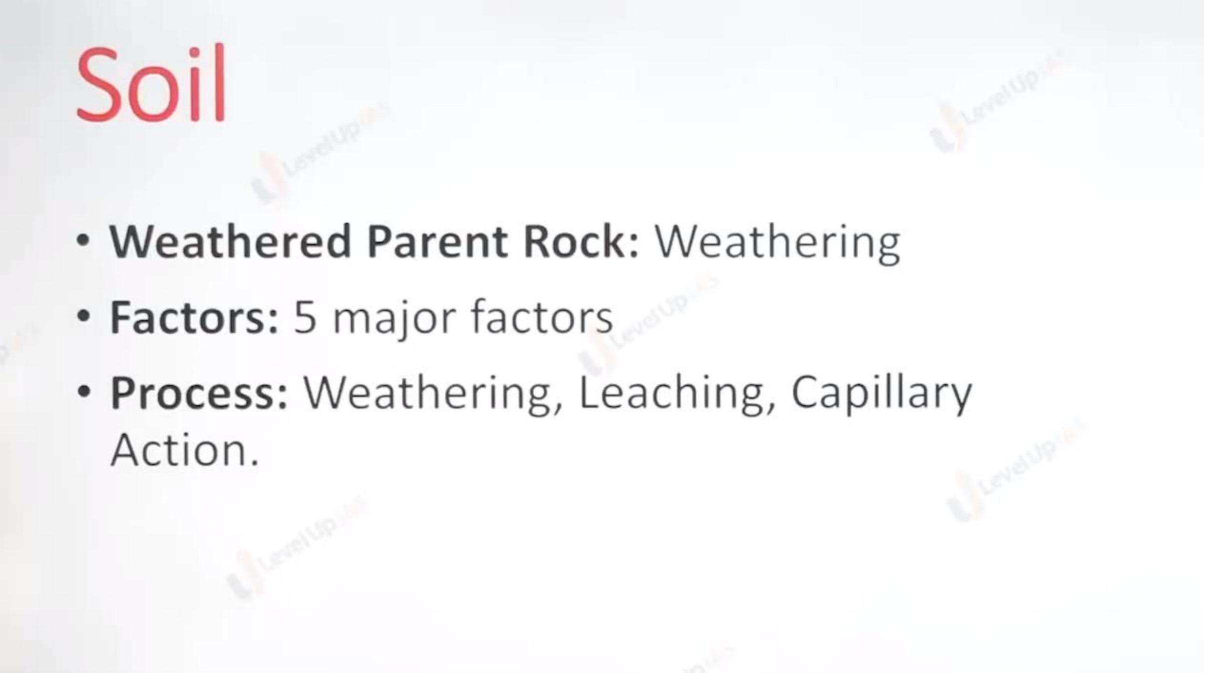




Indian vegetataion contains:

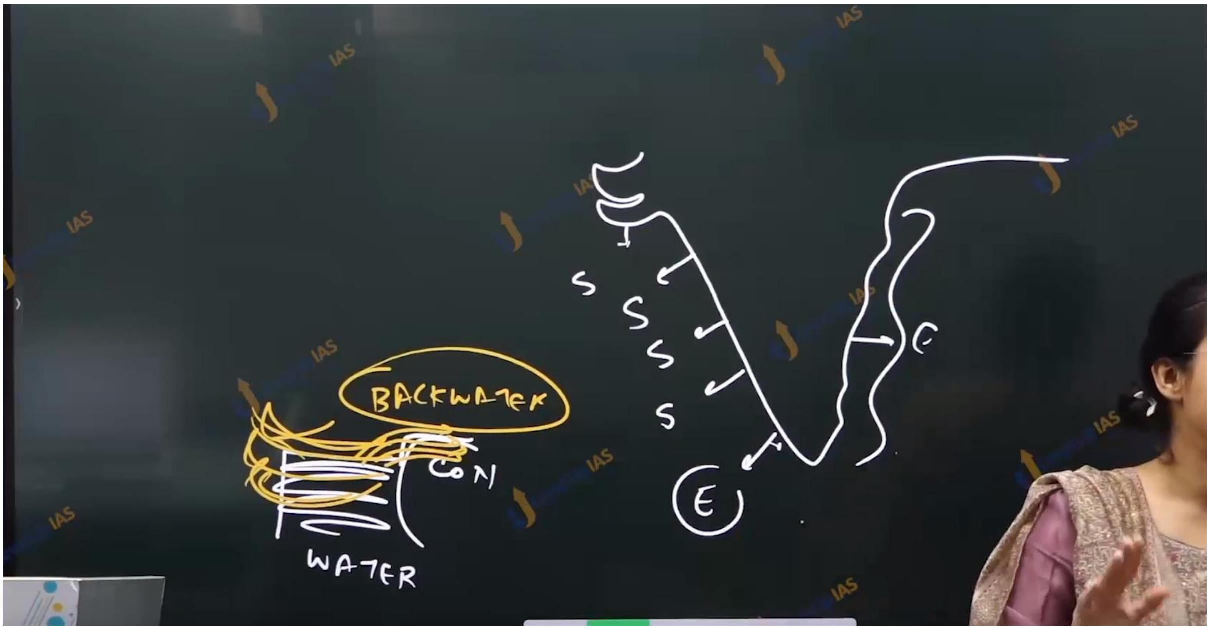
1. Tropical Every green forest:
2. Tropical Deciduous forest:
3. Temperate Decious Forest:
4. Temperate every green forest:
5. Tropical grassland:
6. Temperate grassland:
7. Tundra.:
8. Southern mountain forest:
9. Mangrove
10. Wetlands.
11. Northen montane forest:
12.

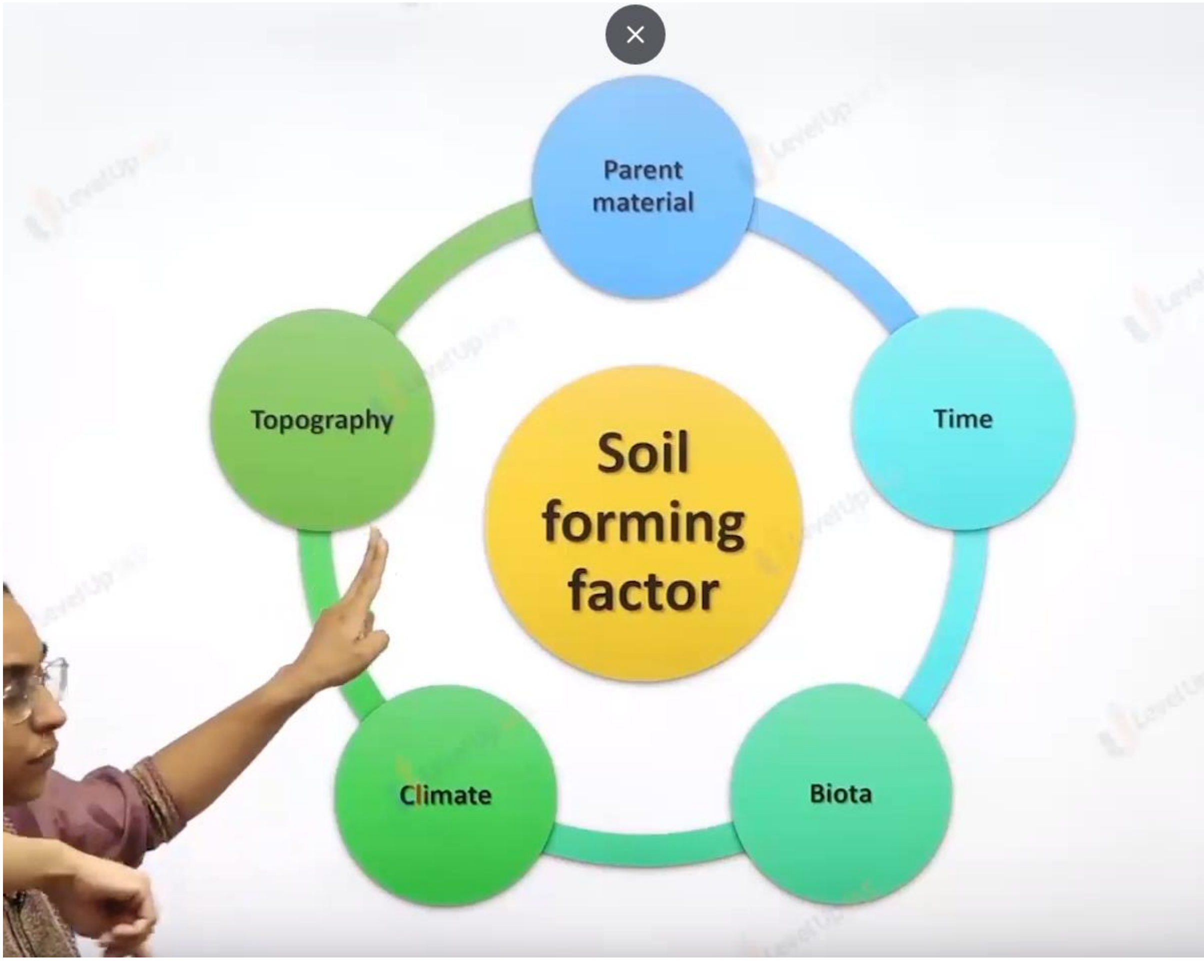




Soil

1. Soil is not simply dirt or broken rock particles but is complex ecosystems in which factor like time, biota , climate play an important role.

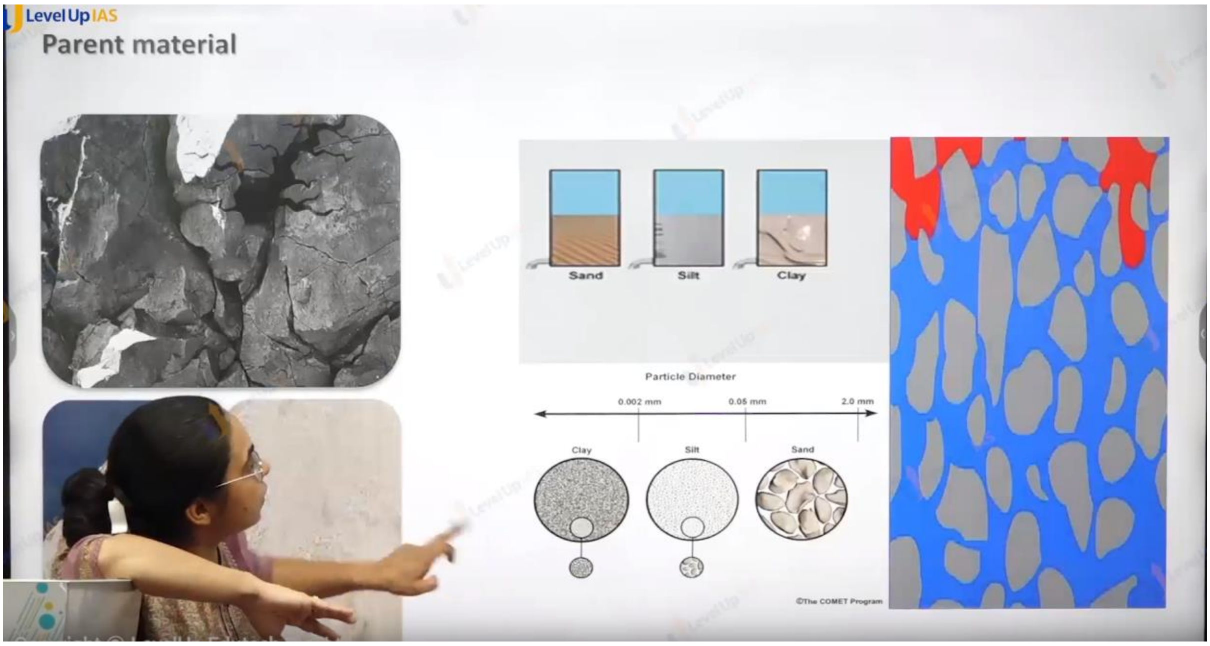


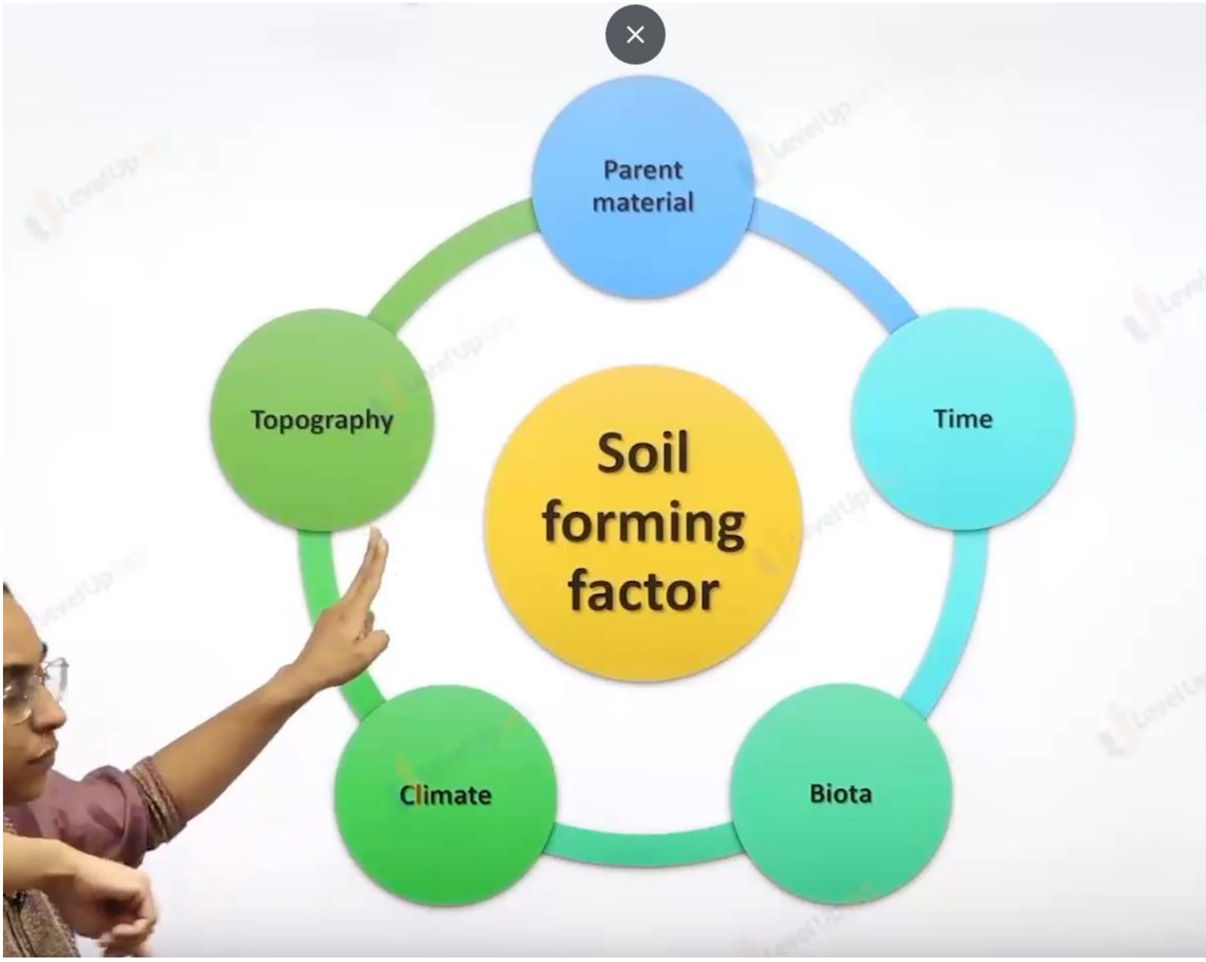


S for submergence

E for emergence.


Level Up IAS
Soil as a Ecosystem
30:29 / 02:30:19


Level Up IAS
Parent material
Sand
Silt
Clay
Particle Diameter
0.002 mm
0.05 mm
2.0 mm
Clay
Silt
Sand
©The COMET Program



Soil Forming Process:

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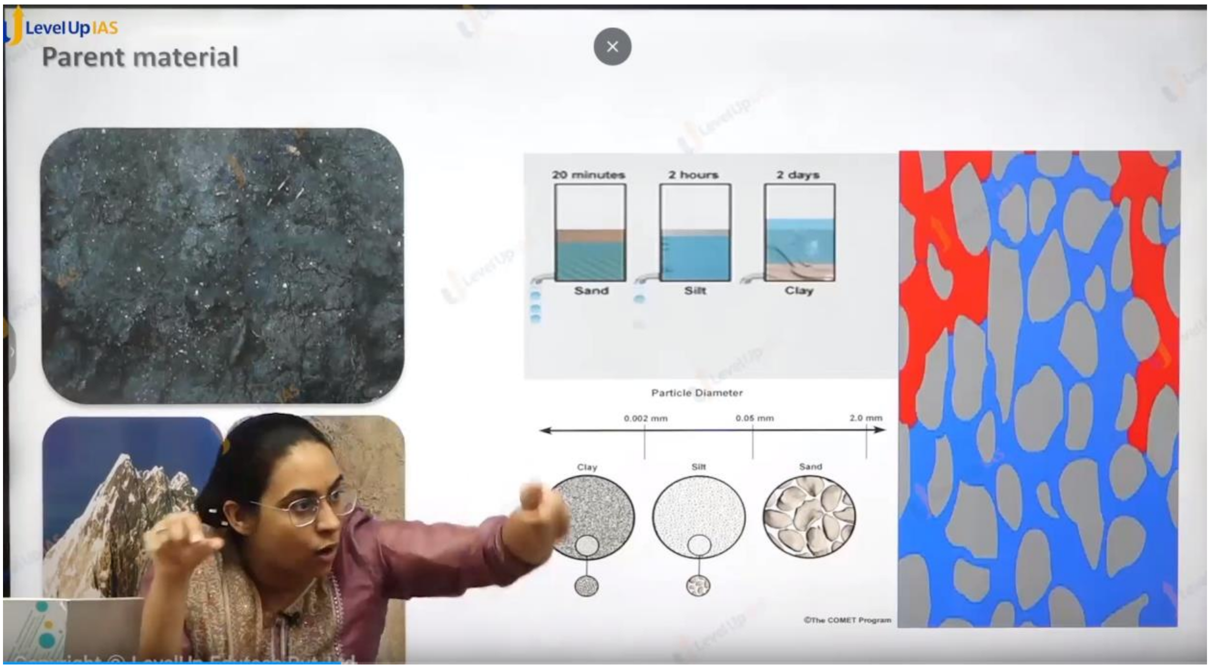
1. Weathering
2. Translocations (water goes down in soil or earth)
3. Leaching (nutrient go down when rains) .
4. Unifaction.(formation of unmoors)
5. Transformation of dirt in the soil.
6.



Parent rock :

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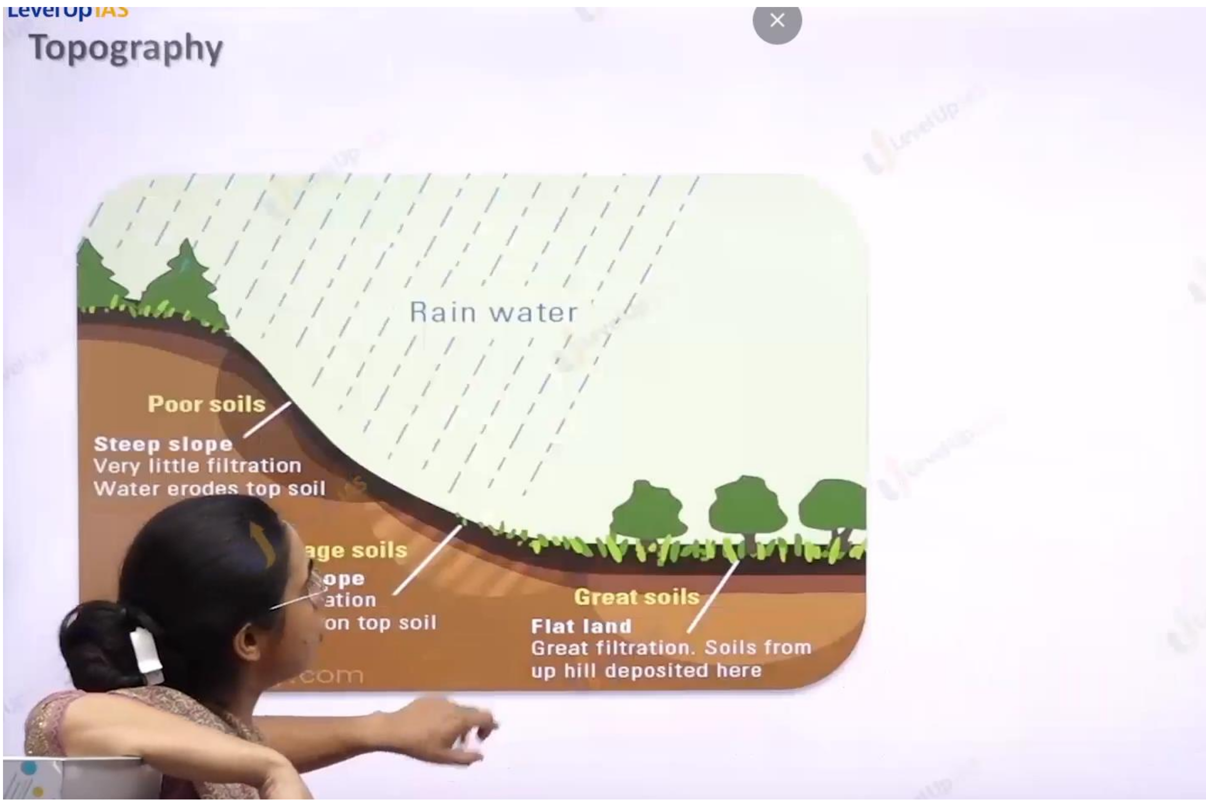
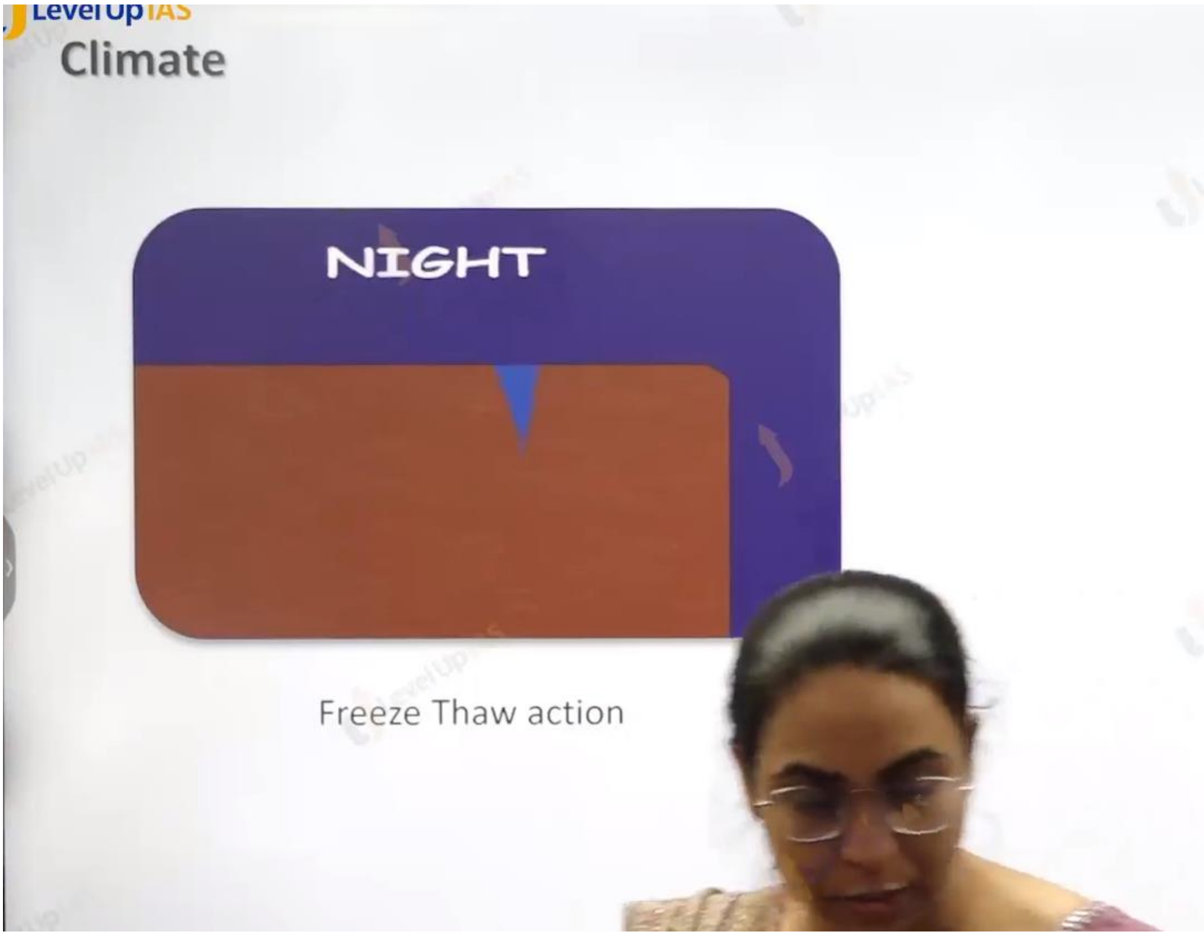
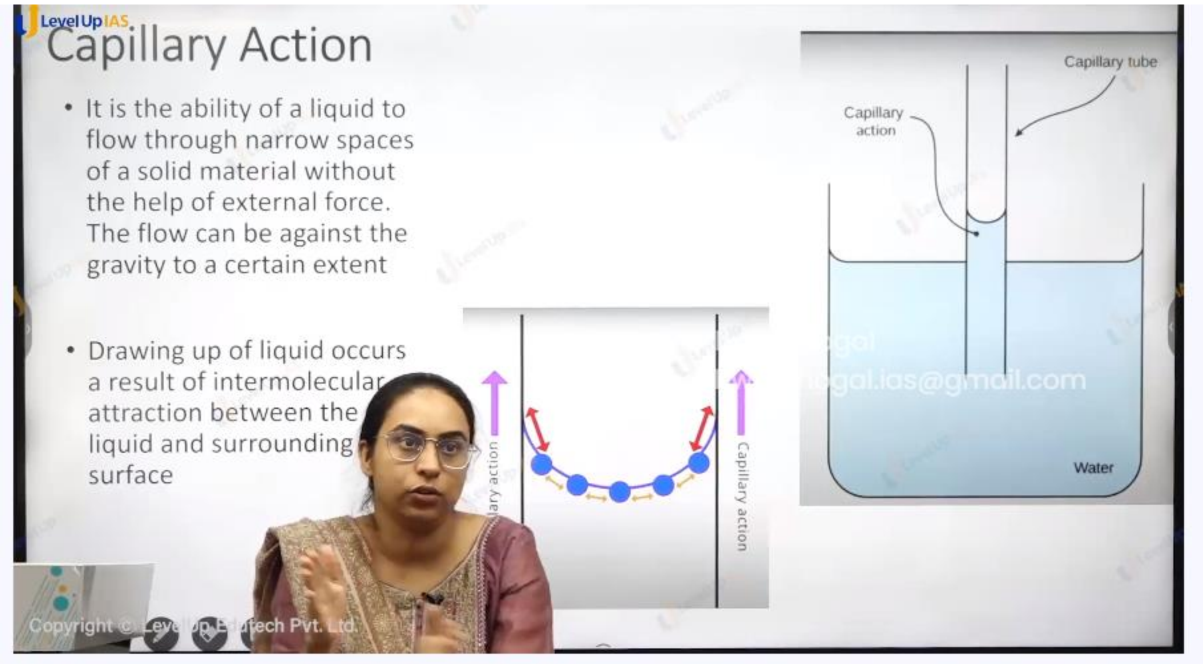
1. Parent rock determine the color of soil. The nutrient, the texture of soil , the permiablity of porosity of soil.
2. Color :
    a. Basalt give black color soil.
    b. Granite gives red color soil.
    c. Limestone goves white color soil.
3. Nutrients:a
    a. Granite give iron rich soil.
    b. Basalt gives soil which is rich in feromagnite siilliecate.
4. Perant rock determines the texture of
    a. Soft rock ex basalt gives fine texture soil which makes soil porous. Ex black cotton soil.
    b. Hard rock gives core texture soil. Which makes the soil permeable. Red soil.
    c.



1. Topography affects the thickness of soil steep slope less thickness of soil gentle slope.


LevelUp IAS
Climate
Freeze Thaw action


LevelUp IAS
Climate
NIGHT
Freeze Thaw action



Climate:

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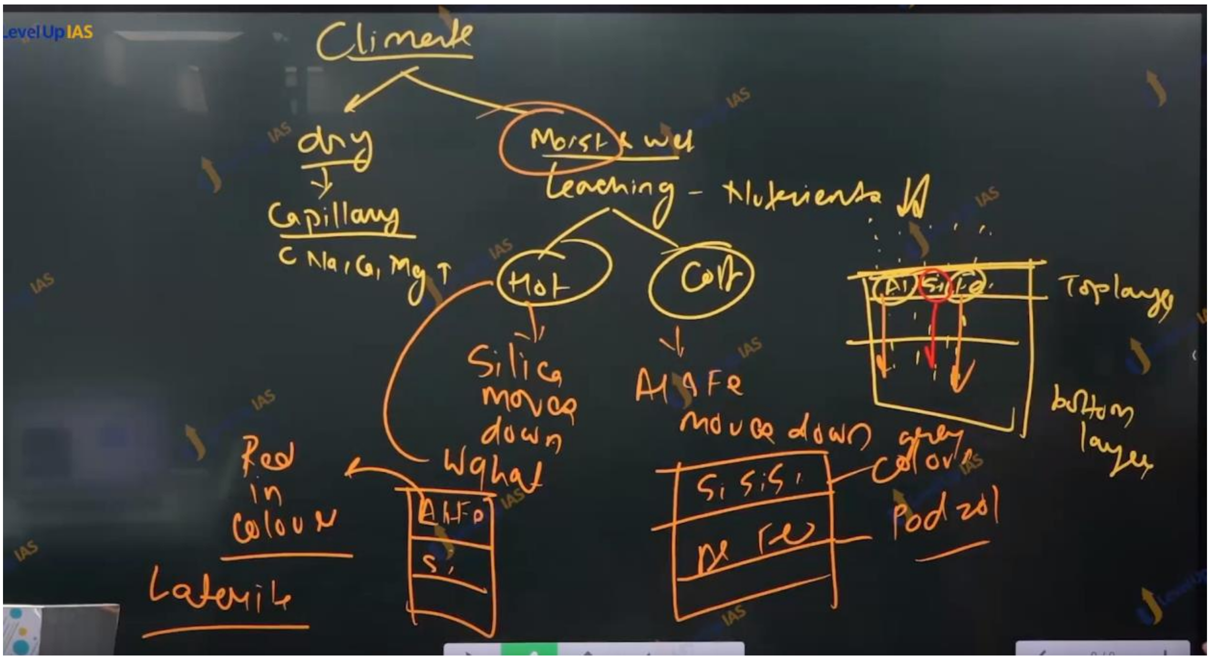
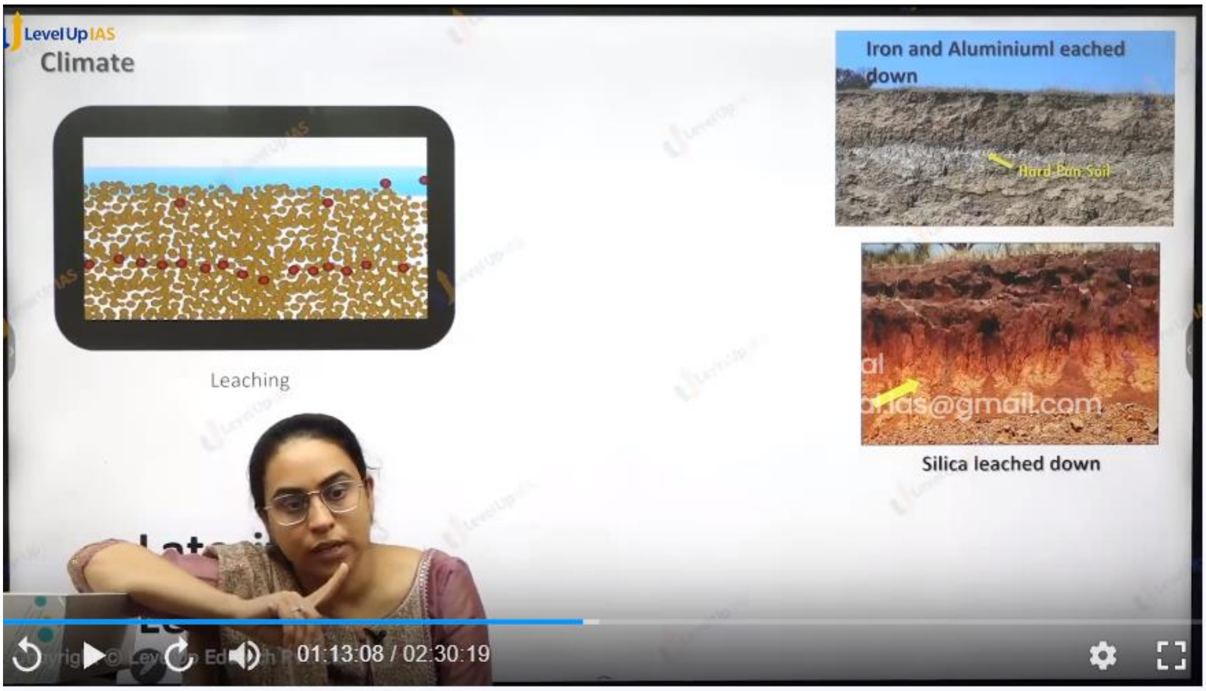
1. Climate determines the process like type of weathering dry climate facilitate the physical weathering whereas moist climate facilitates chemical weathering. Climate also determines whether the process of leaching takes place or process the capillary action takes places.
2. Dry climate facilitates the capillary action whereas humid climate facilitates the leaching
3. Leaching refers to a process where the nutrients are dissolved in the rainwater, and it moves top layer to the lower layers.
4. Leaching very slow process and it generally seen in the area that has more rainfall than evaporations.





Leaching :

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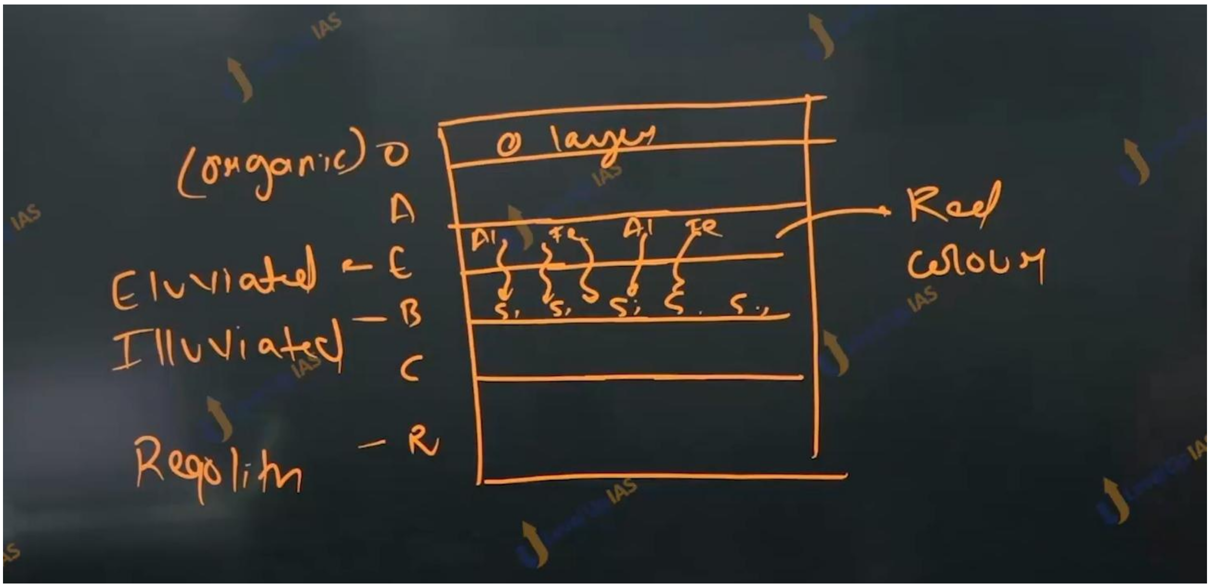
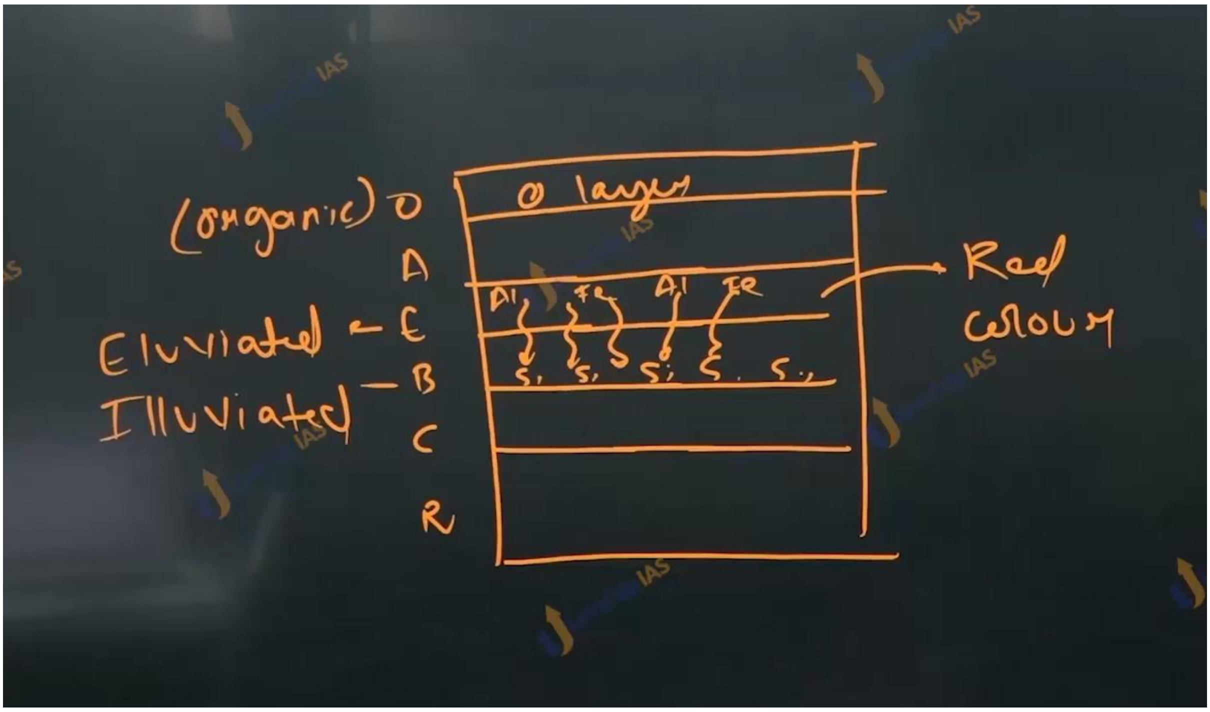
1. Leaching can be of 2 type that is Lateralization and podzolization.

Lateralization:

1. In the region of hot and most conditions silica is leach is down and iron and aluminum oxides are seen in the top layers.
2. Which makes the soil appears red in color such soil is called as lateralized in soil in places that has alternate wet and dry conditions. Such as it seen in western ghat and eastern ghat.
3. Laterization soil is used for brick making it also support of forestry and some type of plantation of agricultural cultivation of rubber coffee tea.
4. Unlike the red soil which is red in color due to parent material lateraled soil is red in color due to the process of leaching.





Podzolization:

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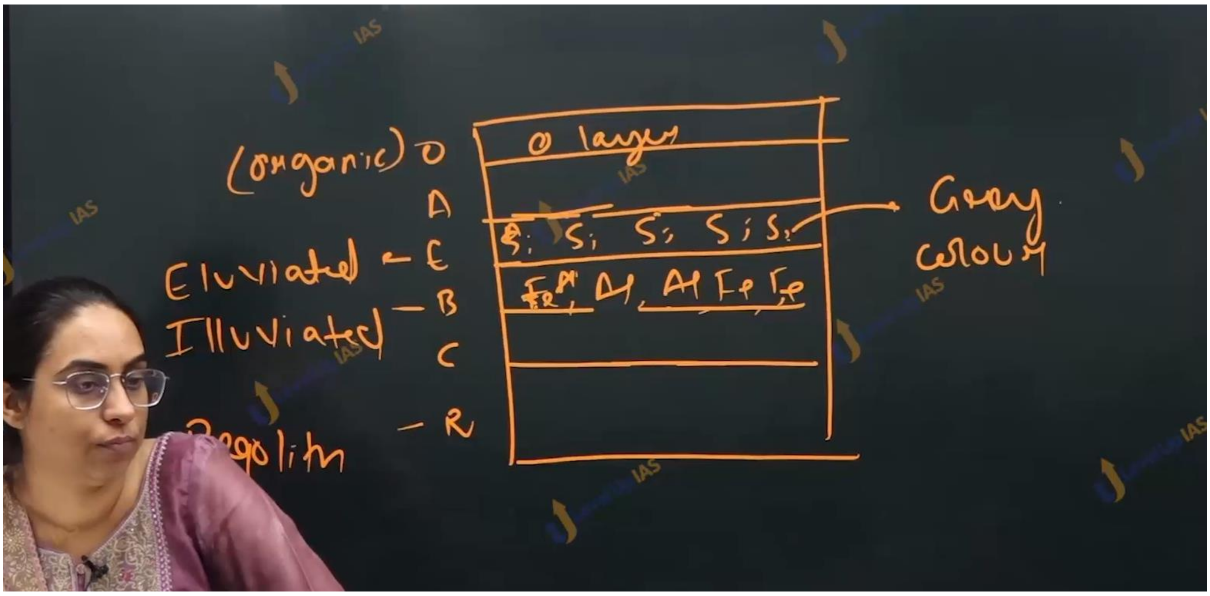
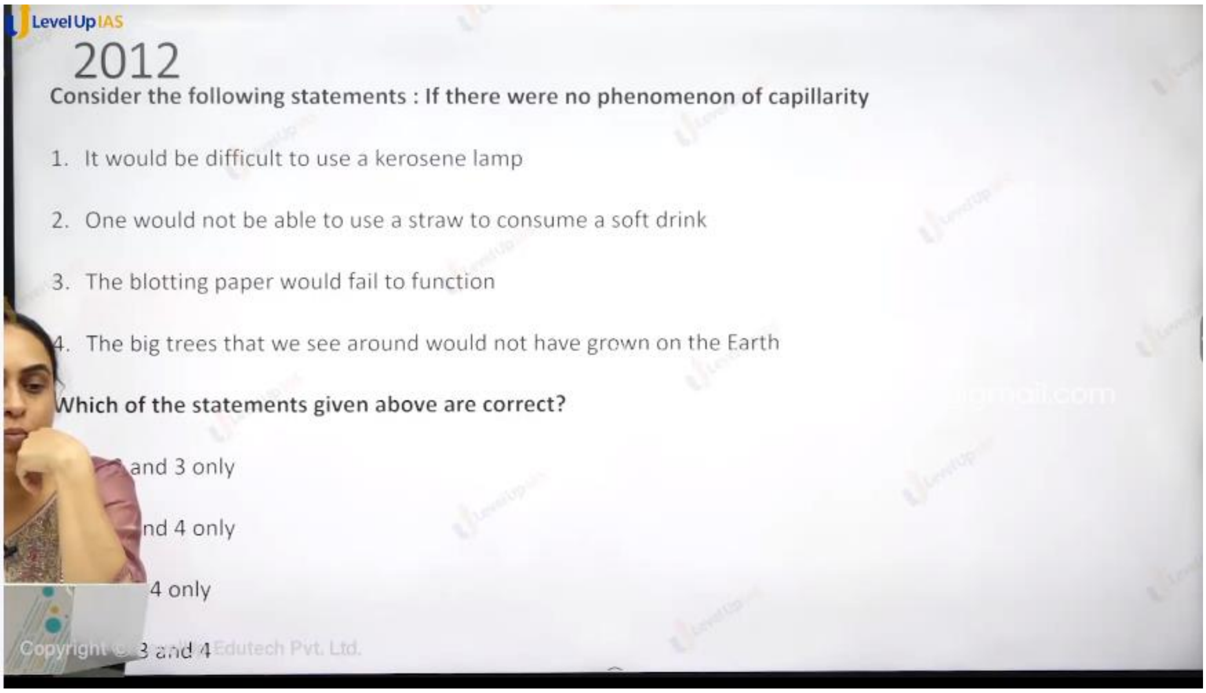
1. In the area of cold and humid conditions, iron and aluminum is leached whereas silica remains int eh top layer this makes the soil. Gray in color. It is seen in India in the top reaches of Himalayas
2. Podzolic is generally seen in temperate regions.

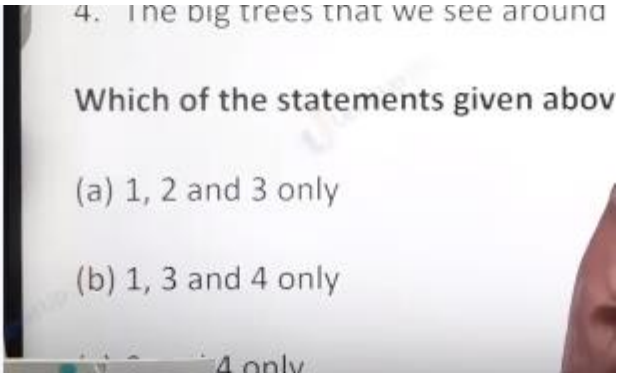




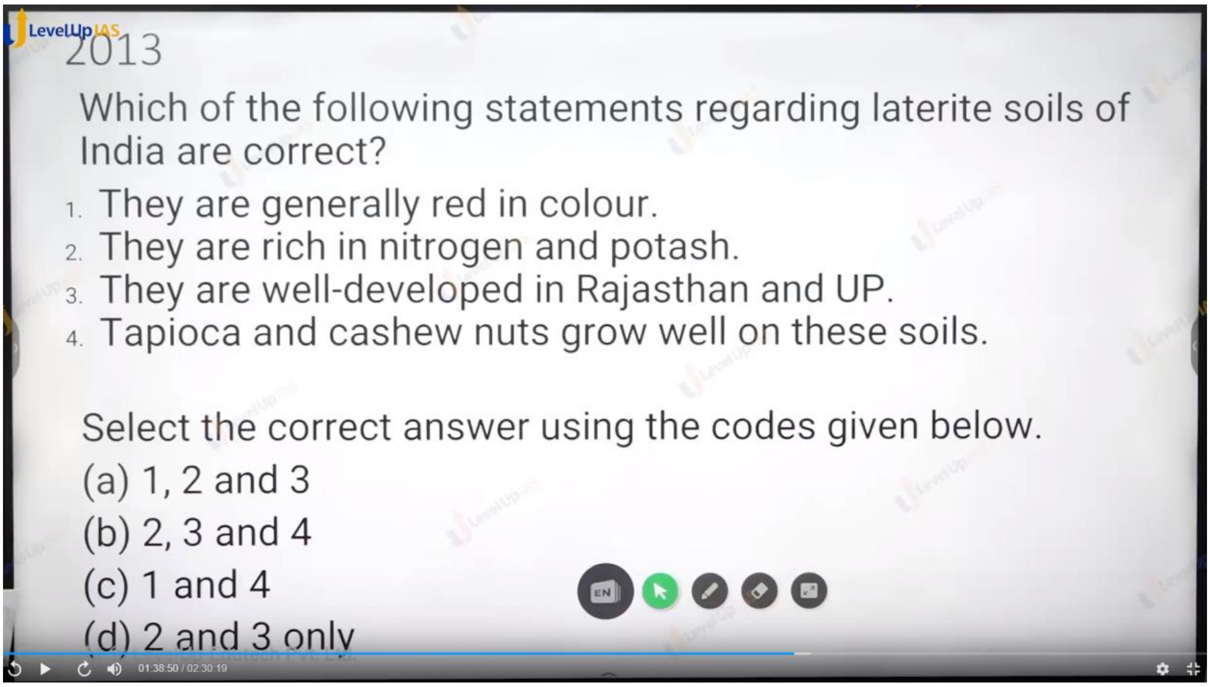
Options are not visible.

Option is c 3 and 4

Option d is all of above.

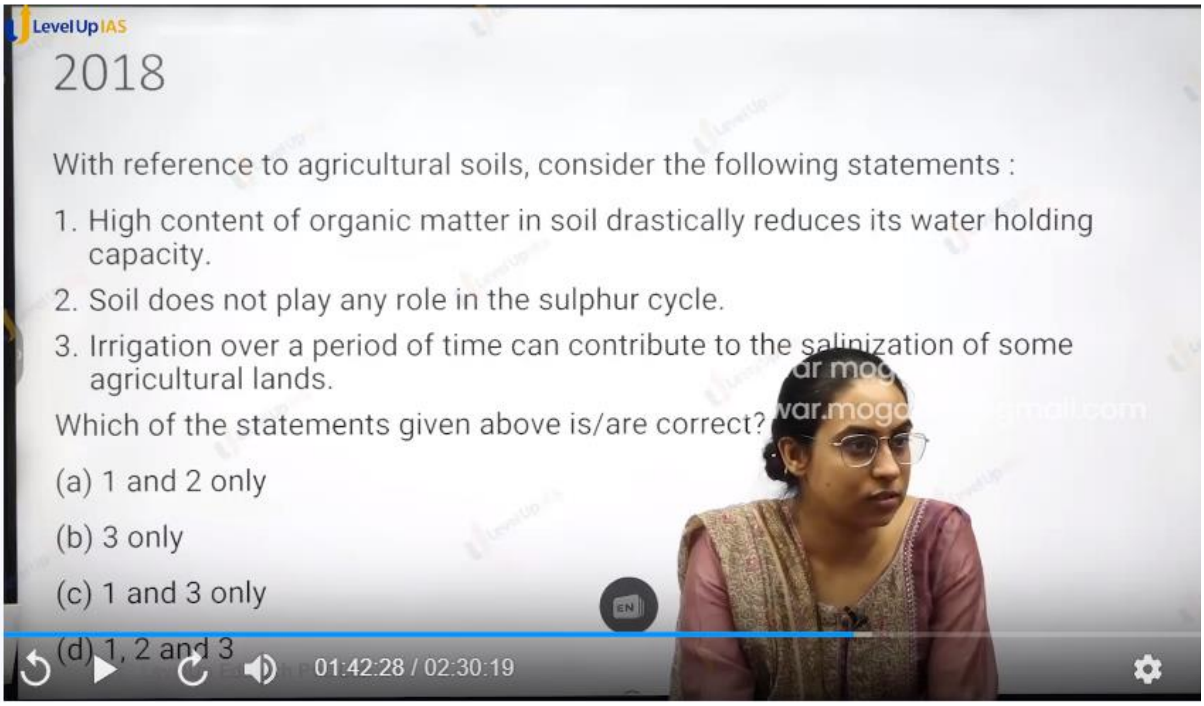


Ans is b.

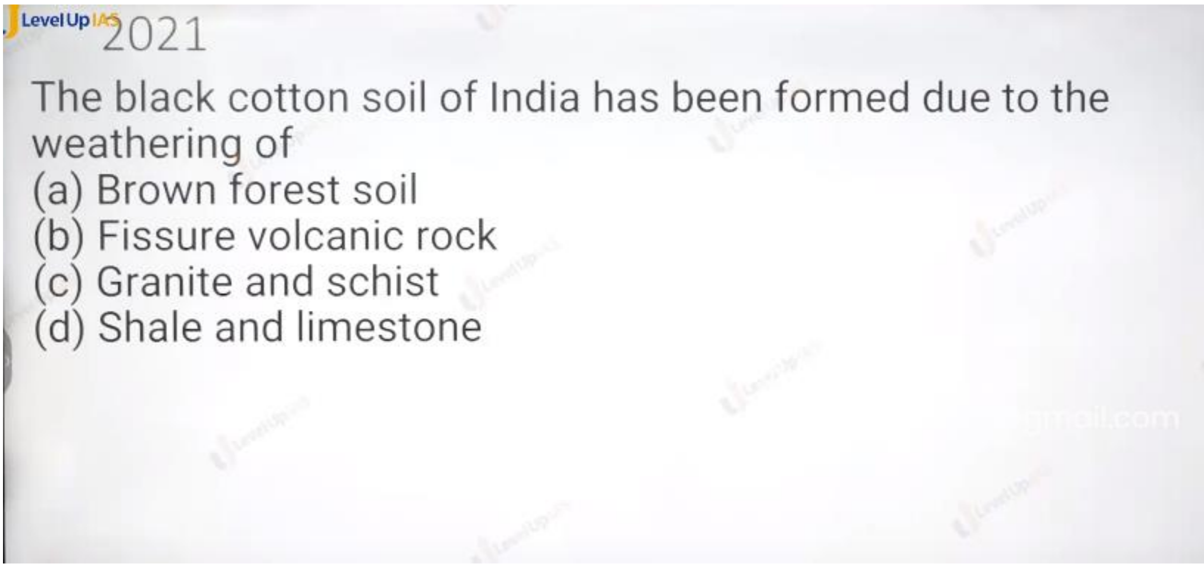


Ans is c.

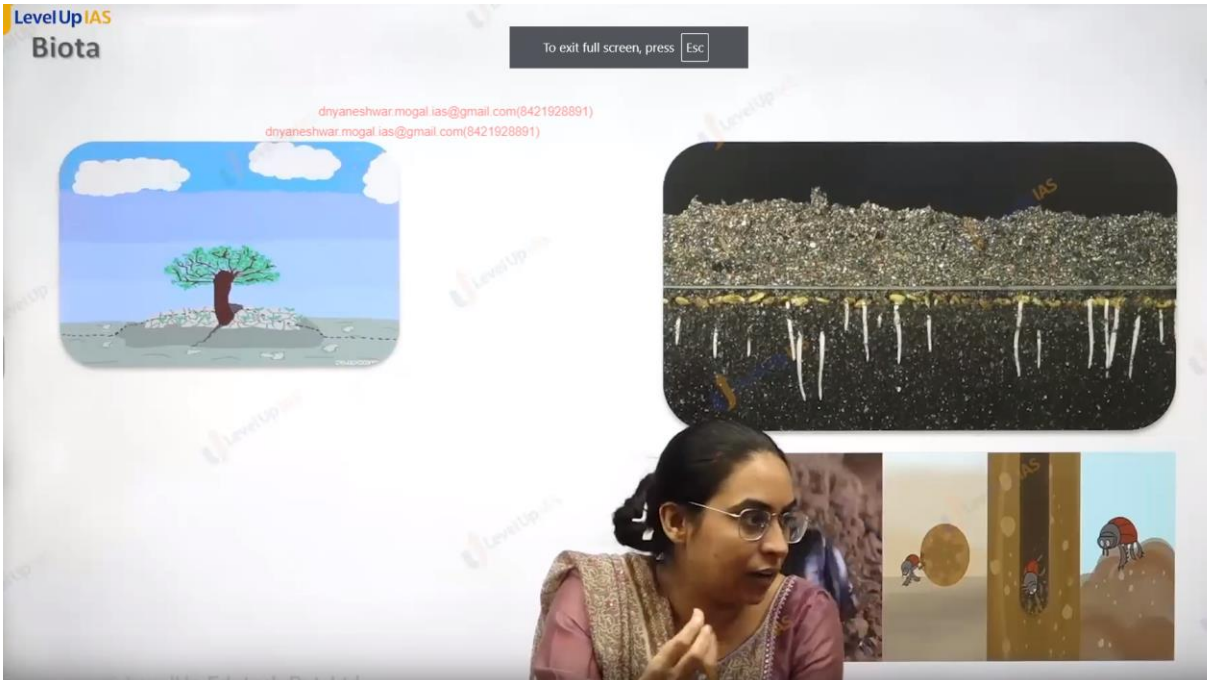
r



ans id b 3 only



Ans is b : granite red soil.



Biota:

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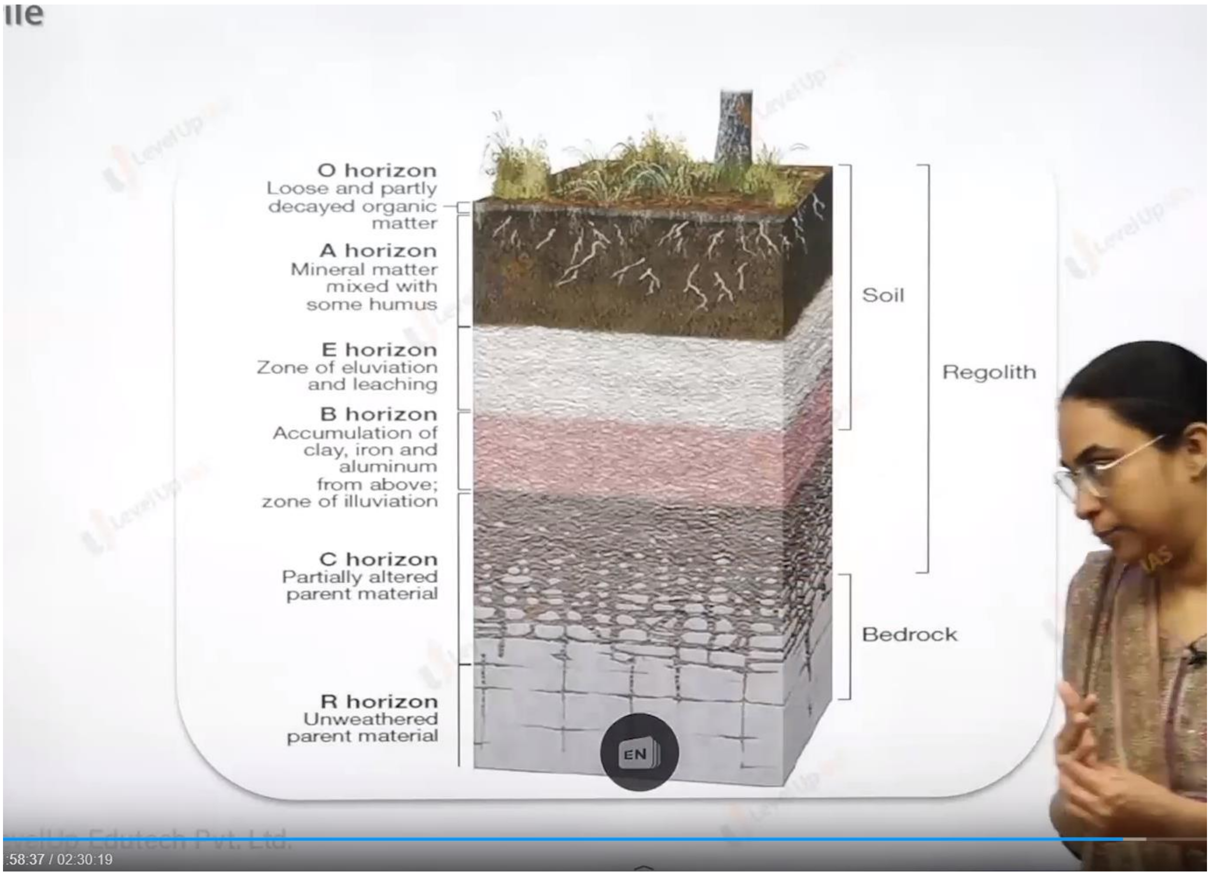
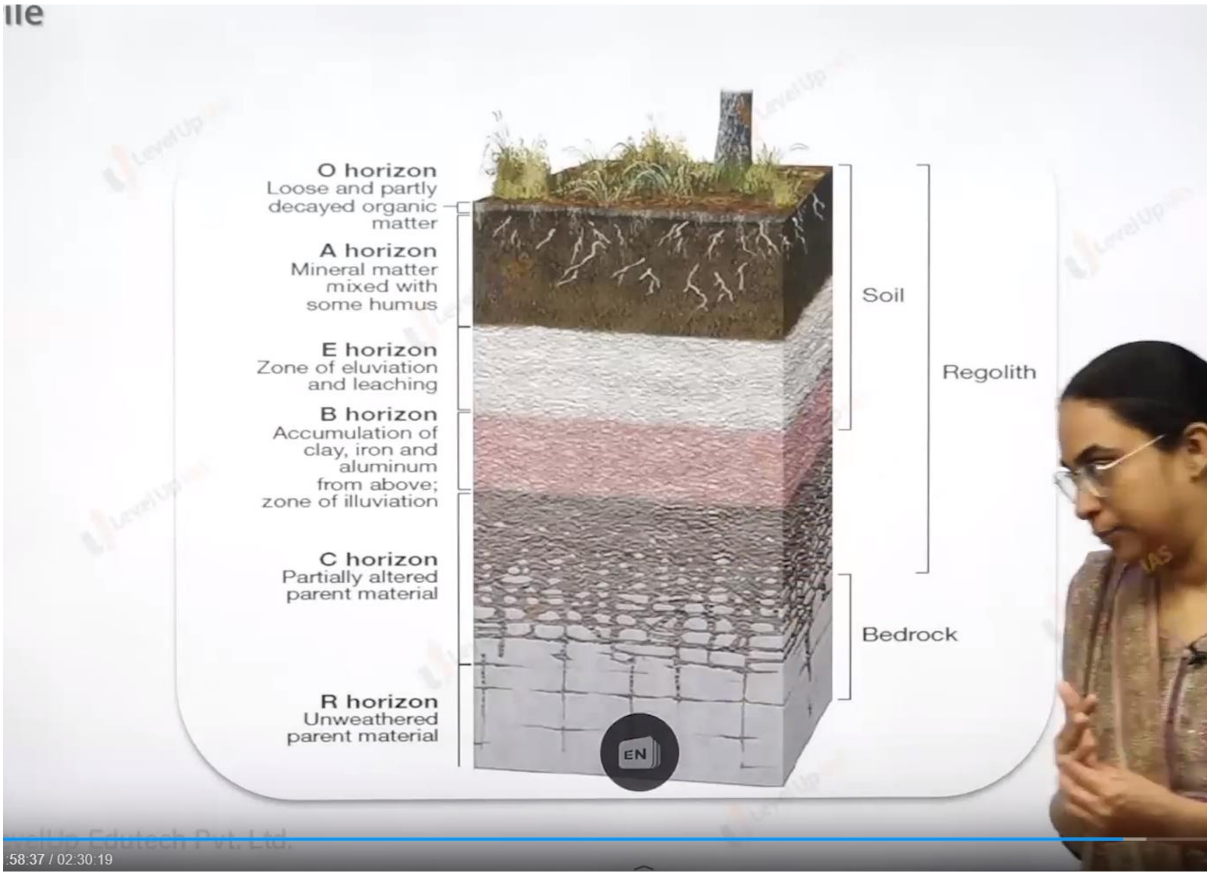
1. The tree breaks the rock and leads to weathering. Earthworm can turn the soil. And helps in bio terbations.
2. The accumulation of humas because the decomposition of leaves steam shrubs etc.

3. Whether humas keep accumulating or would be decomposed depends on bacterial acidity. In the region high temperatures high bacterial activity leads to disintegration of humas containing to less fertility of soil.
4. Whereas in the temperate region more acclamation of humas soil is very fertile. Ex. the soil in temperate grasslands. Steeps black sea.
5.


O horizon
Loose and partly decayed organic matter
A horizon
Mineral matter mixed with some humus
E horizon
Zone of eluviation and leaching
B horizon
Accumulation of clay, iron and aluminum from above; zone of illuviation
C horizon
Partially altered parent material
R horizon
Unweathered parent material
Soil
Regolith
Bedrock
EN
58:37 / 02:30:19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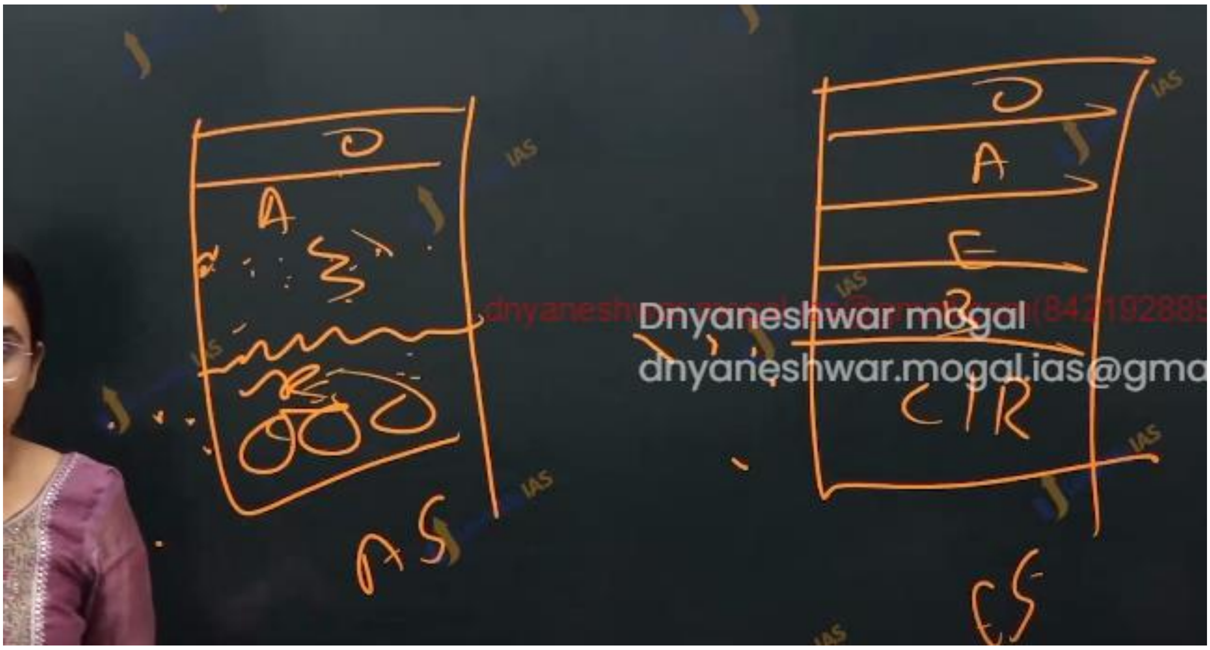
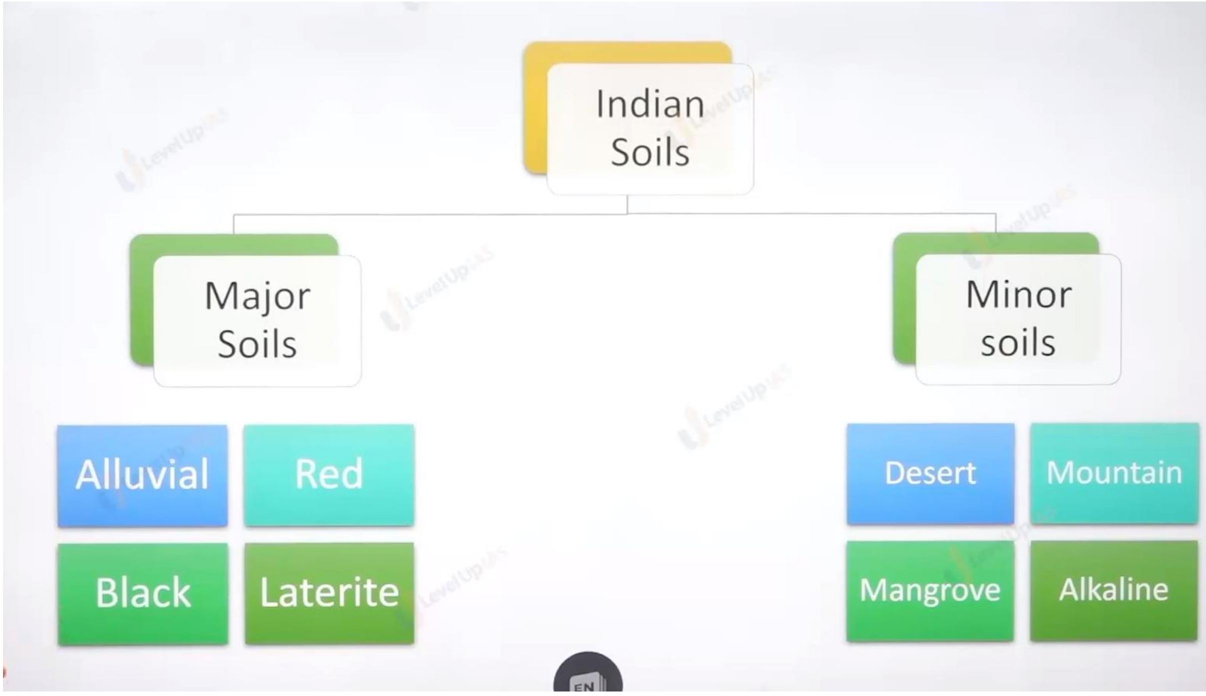
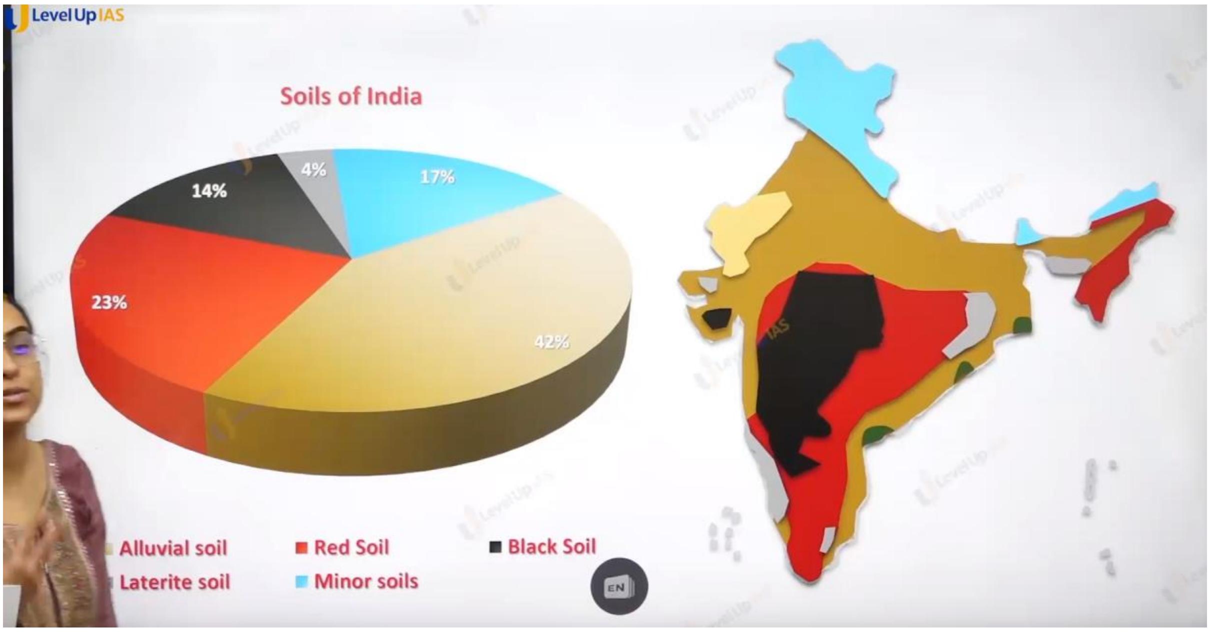
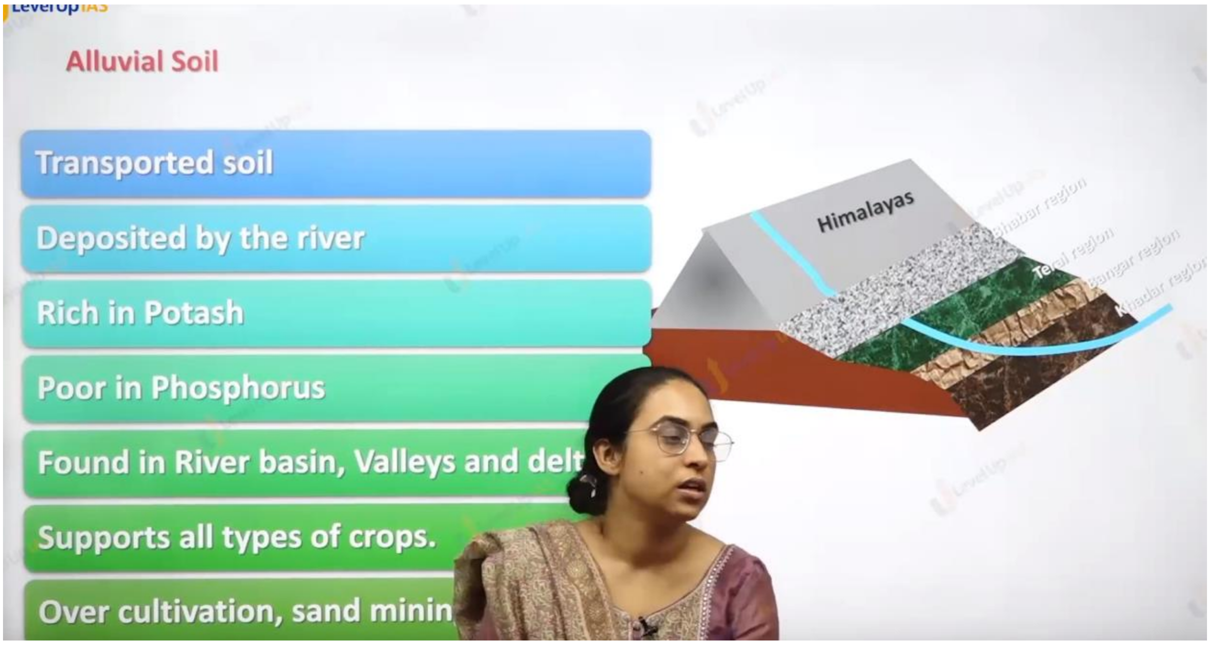


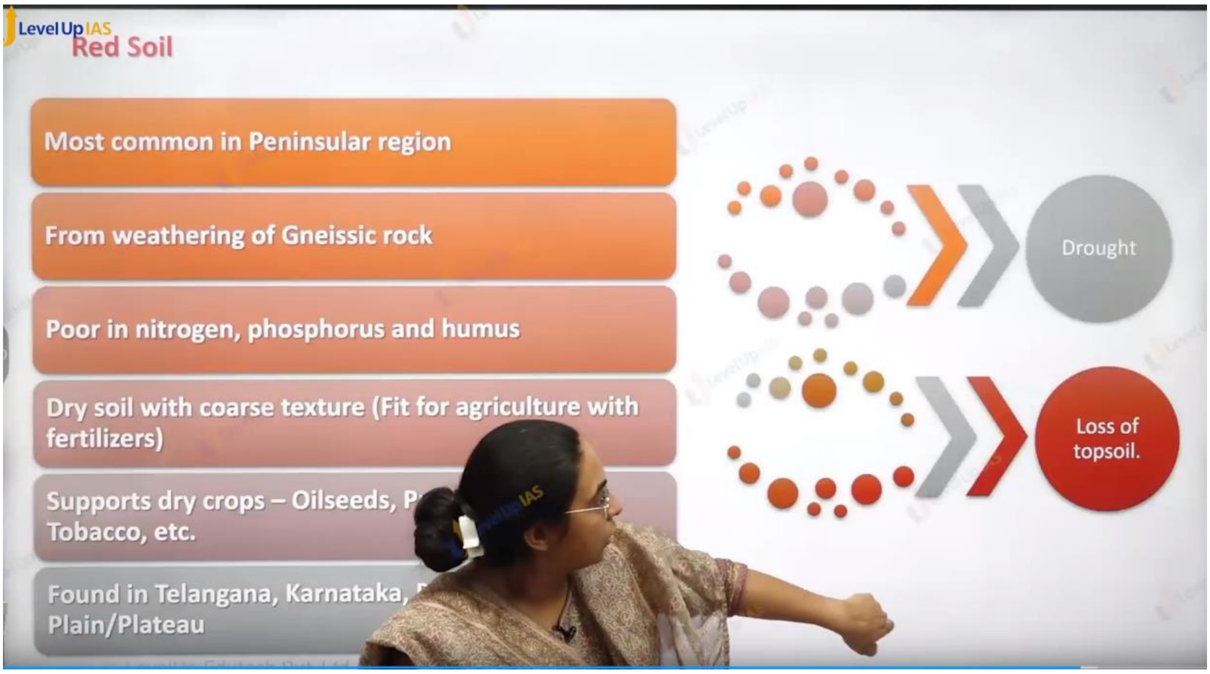
Role of Time

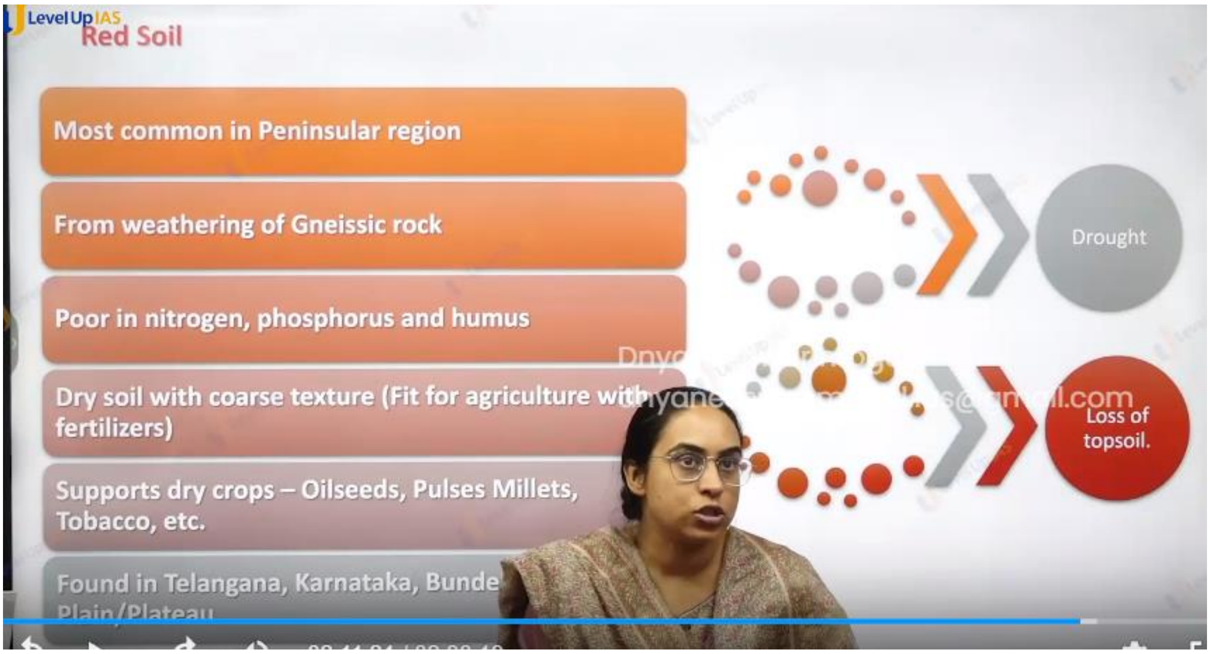
1. Times determines whether the soil has developed many horizons or not. Time determines the mutuality of soil.
2. Mature soil Has fully developed horizontally. Ex. Equatorial soil. Whereas young soil does not have fully developed horizons.




LevelUp IAS
Soils of India
4%
17%
14%
23%
42%
Alluvial soil
Red Soil
Black Soil
Laterite soil
Minor soils
EN


Alluvial Soil
Transported soil
Deposited by the river
Rich in Potash
Poor in Phosphorus
Found in River basin, Valleys and delt
Supports all types of crops.
Over cultivation, sand minin
Himalayas
Bhabar region
Terai region
Bangar region
Khadar region


Level Up IAS
Red Soil
Most common in Peninsular region
From weathering of Gneissic rock
Poor in nitrogen, phosphorus and humus
Dry soil with coarse texture (Fit for agriculture with fertilizers)
Supports dry crops – Oilseeds,
Tobacco, etc.
Found in Telangana, Karnataka,
Plain/Plateau
Drought
Loss of topsoil.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Found in Telangana, Karnataka, Bunde
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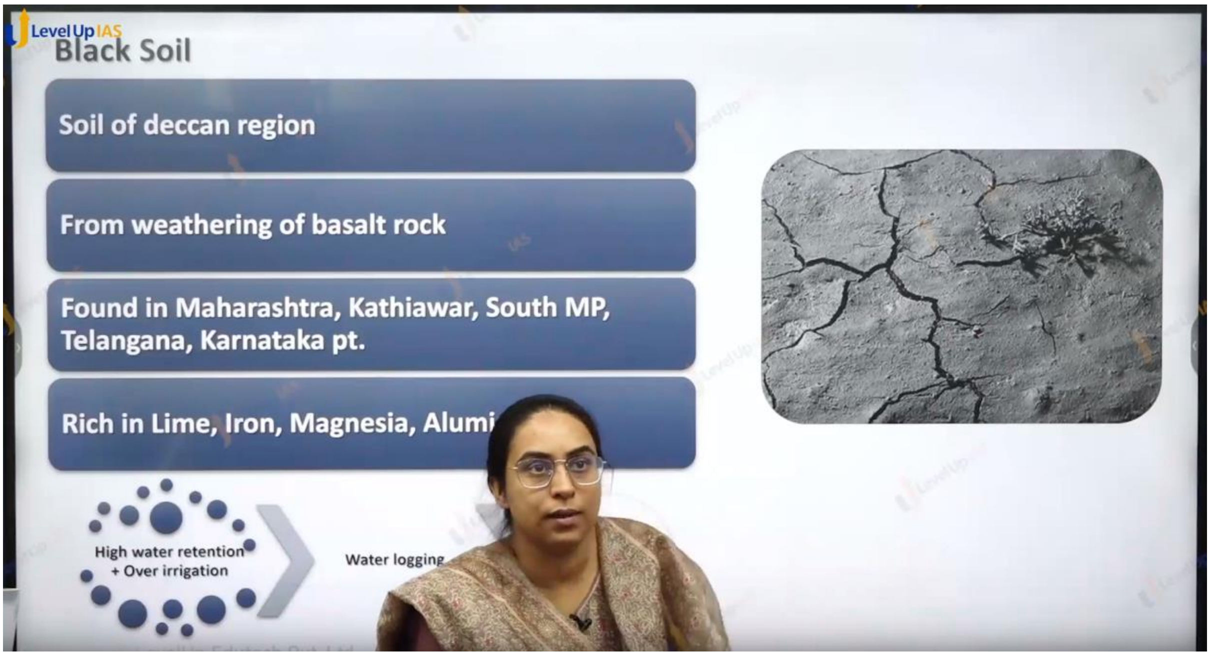
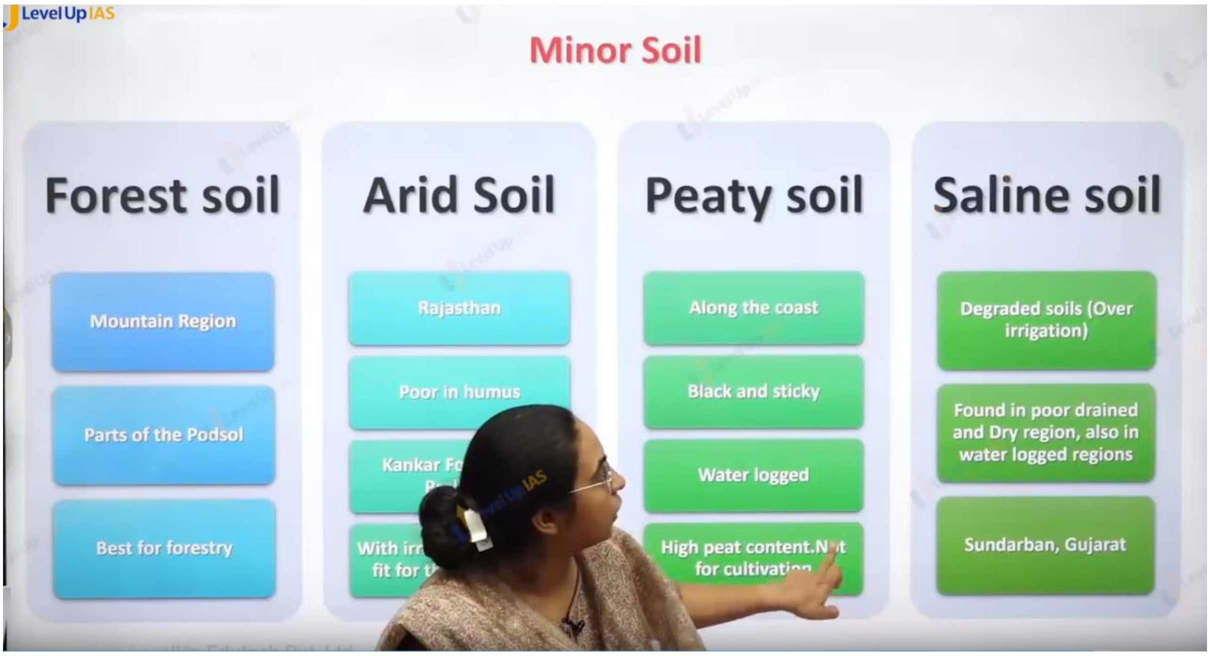
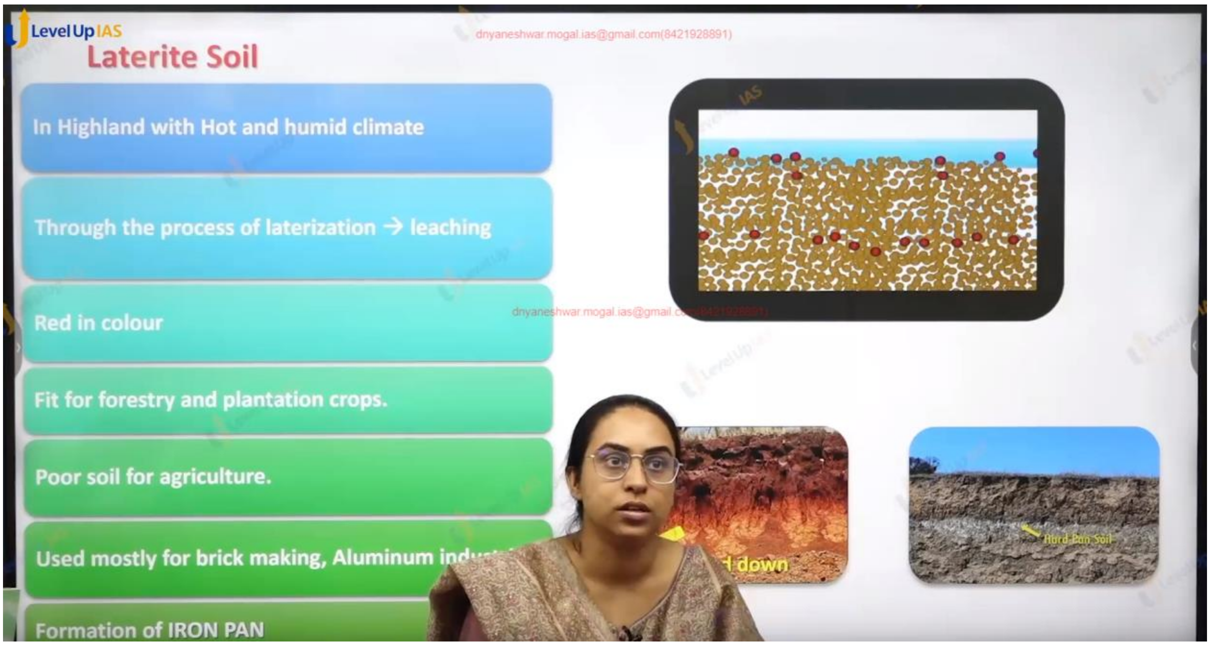
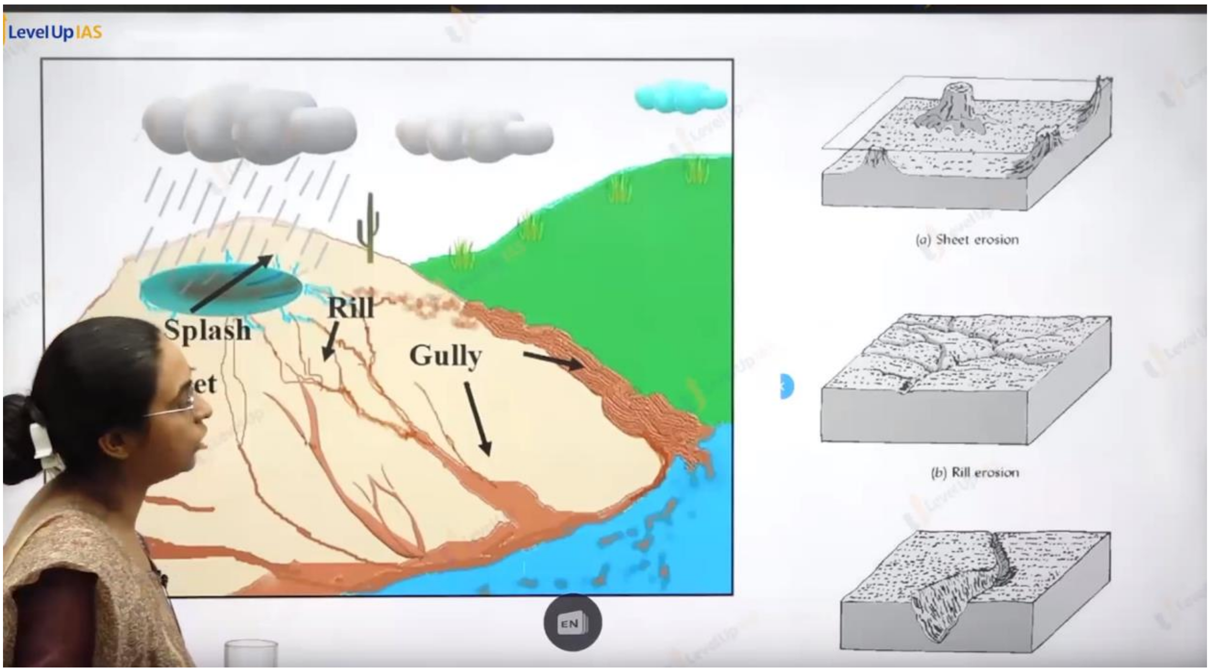

LevelUp IAS
Black Soil
Soil of deccan region
From weathering of basalt rock
Found in Maharashtra, Kathiawar, South MP, Telangana, Karnataka pt.
Rich in Lime, Iron, Magnesia, Alumi
High water retention + Over irrigation
Water logging


LevelUp IAS
Minor Soil
Forest soil
Mountain Region
Parts of the Podsol
Best for forestry
Arid Soil
Rajasthan
Poor in humus
Peaty soil
Along the coast
Black and sticky
Water logged
Saline soil
Degraded soils (Over irrigation)
Found in poor drained and Dry region, also in water logged regions
Sundarban, Gujarat

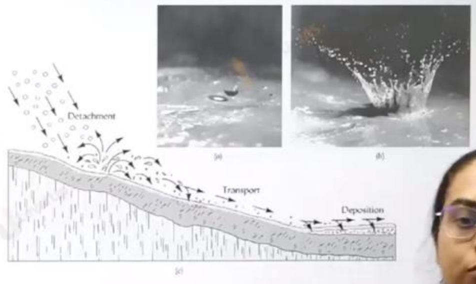

Level Up IAS
Laterite Soil
dnyaneshwar.mogal.ias@gmail.com(8421928891)
In Highland with Hot and humid climate
Through the process of laterization → leaching
Red in colour
Fit for forestry and plantation crops.
Poor soil for agriculture.
Used mostly for brick making, Aluminum indu
Formation of IRON PAN
Hard Pan Soil


Level Up IAS
Splash
Rill
Gully
(a) Sheet erosion
(b) Rill erosion

SOIL PROBLEMS

1. Soil Erosion:

- Removal of fertile top layer. It is associated with mining, wrong agricultural practices like fallow land (bare land without any crop cover), deforestation, over grazing.
- Fallowing of land exposes soil to hitting of raindrop leading to splash erosion. Continuous erosion by raindrop leads to erosion of entire surface in form of sheet known as sheet erosion.
- Surface water flows that cause sheet erosion rarely travel more than a few metres before turning into rills and causing Rill Erosion
- Rill eventually widen to become gully erosion



1 (a) Raindrop falling on the surface (b) Splash impact of raindrop (c) Process of water erosion (modified ... Stitcher 2010) 12.2.1.3. Rill erosion (channel erosion)

SOIL PROBLEMS

2. Soil Degradation: It refers to decline in productivity and quality of soil. It can be due to:

- **Soil Pollution:** Excessive use of chemical fertilizers, pesiticides etc Waste Dumping, Mining, Disposal of industrial effluents
- **Wrong Agricultural Practices** like monoculture, unscientific rotation of crops, salinisation due to cultivation of water intensive crops in dry regions, Shifting Cultivation.
- Capillary action leads to accumulation of calcium carbonate beneath (kankar), the soil will act as impermeable to water and water gets logged as happened in Indira Gandhi canal regions of Rajasthan.

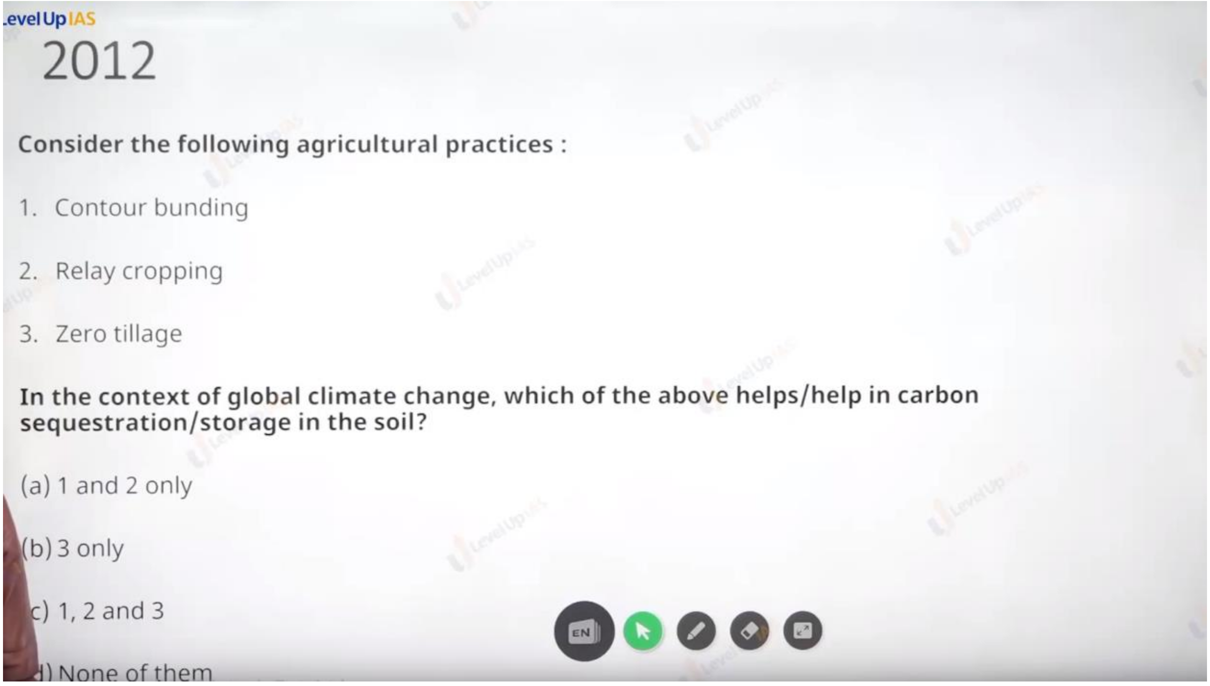
SOIL PROBLEMS

3. Soil Desertification: Extreme form of soil degradation where soil is no longer fit for cultivation

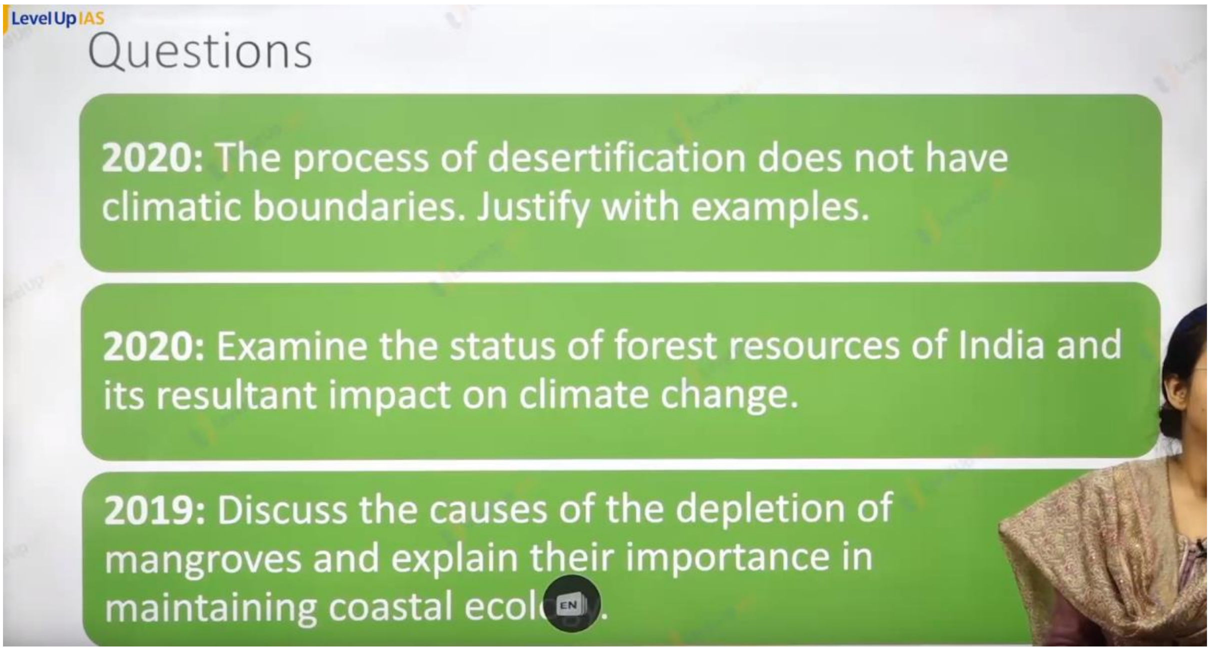
- **Reasons:**
  - Spread of desert like conditions along the productive soil.
  - Extreme form of Land degradation: Same points as degradation but with more intensity

Solutions

- **Soil organic matter** should be maintained as it provides fertility, better water retention and holds the soil together
- **Plant trees, shrubs** to provide wind protection and also binds soil to roots.
- **Mulching:** Covering on ground of hay, straw and crops residue which prevents soil from erosion and preserves moisture in soil. It also prevents growth of weeds
- **No-Till/Minimum Tillage Methods** to conserve the soil of moisture, reduce erosion and does not disturb the soil
- **Erosion-reducing grazing practises:** Rotational grazing is a method of moving livestock from one pasture paddock to the next. Each paddock is given a rest period and is allowed to regrow naturally, reducing soil compaction and erosion.
- **Terracing esp in hilly areas** to prevent ploughing against the slope.
- **Windbreaks (also known as shelterbelts)** are rows of trees and shrubs planted along the edges of agricultural fields to provide wind protection
- **Organic farming, Precision farming** to reduce the excessive use of fertilizers, pesticide
- **Growing climate suitable crops** to prevent soil degradation


LevelUp IAS
2012
Consider the following agricultural practices :
1. Contour bunding
2. Relay cropping
3. Zero tillage
In the context of global climate change, which of the above helps/help in carbon sequestration/storage in the soil?
(a) 1 and 2 only
(b) 3 only
c) 1, 2 and 3
d) None of them

Ans c .


LevelUp IAS
Questions
2020: The process of desertification does not have climatic boundaries. Justify with examples.
2020: Examine the status of forest resources of India and its resultant impact on climate change.
2019: Discuss the causes of the depletion of mangroves and explain their importance in maintaining coastal ecol